Striking Departure from Polygenic Architecture in the Tails of Complex Traits

immediate

Abstract

Understanding the genetic architecture of human traits is a key goal of biomedical research and precision medicine. However, little is known about how genetic architecture varies across the trait continuum and, in particular, if it differs in the tails of complex traits, where disease often occurs. Here we show, via simulation, that trait tails are enriched for rare alleles of large effect under stabilising selection, a key factor shaping genetic variation in humans. Thus, a substantial fraction of individuals may have extreme trait values due to only one or several alleles of very large effect, rather than polygenic burden. To test this, we introduce a method that utilizes polygenic risk scores (PRS) to detect departures from polygenic architecture in trait tails from population data: the PRS-On-Phenotype outlier ("POPout") test. Applying POPout to UK Biobank (UKB) data, we observe significant - often striking - evidence of departures in polygenic architecture in 107 of 148 complex trait tails tested. These results replicate in non-European UKB individuals (R=0.63; P=1x10-5), in the US-based All Of Us dataset (R=0.59; P=1x10-4), in individuals with repeat measures (R=0.71; P=1x10-8), and using our sibling-based test of tail architecture (R=0.81; P=1x10-9). We demonstrate that PRSs perform relatively poorly in tails with reduced polygenic architecture and that those tails are associated with enrichment of rare alleles (R=0.34; P=1x10-5) and advanced paternal age (P=0.004). Finally, we find evidence of ongoing selection consistent with observed departures in polygenicity. Overall, our findings suggest that selection drives rare, large-effect alleles to the tails of complex traits, where, consequently, disease often manifests. Our study has implications for the discovery of rare variants, for the utility of polygenic scores and for the relative importance of common and rare variants to human traits and disease.

Introduction

Complex polygenic traits are typically modeled as the additive sum of many variants of limited effect alongside rare risk alleles of substantial penetrence. This distinction likely reflects a balance between long-range selection pressure and the limited ability of trait-altering mutant alleles to persist over multiple generations. Identification of individuals with extreme trait values as probable carriers of rare high-impact variants has led sequencing strategies to focus on individuals in the tails of the trait distribution. Tail trait values can also result from extreme polygenic burden created by recombination and reassortment. Therefore, to enhance the efficency of sequencing studies, it is imperative to be able to discriminate between these different etiologies.

One indicatation that rare variants may play a significant role in the tails of the trait distribution can be reflected in the reliability of polygenic risk scores (PRS) in the tails. If highly penetrant rare variants not included in GWAS determine the tails of the trait distribution such individuals will not be assigned appropriate risk scores. In this study, we develop a *polygenic regression-to-mean* (PRM) statistical test that compares the PRS in the tails of the trait distribution to expectation PRS under a polygenic model. Applying this test to hundreds of traits in the UK Biobank reveals that the majority of quantitive traits exhibit significant PRM in one or both tails.

Next we examine whether rare variants in exome studies are enriched in trait tails with significant tail regression and demonstrate that such tails are more likely to be under negative selection. Further examining these trait tails we find an association between PRM and other measures of fitness that may be influenced by rare variants including the lifetime number of illnesses and still births. We then compare these results to a simulation and find that they match.

Finally, we assess whether the predictions of a complex trait architecture derived from a conditional sibling model align with the observed diversity in the architecture of trait extremes. The concordance between these measures underscores the

need to consider the role of rare variants in the analysis of trait tails. The PRS and sibling analyses presented here offer valuable tools for the targeted selection of traits and samples likely to harbor de novo or rare deleterious alleles and represent an approach that has the potential to significantly enhance the effectiveness and design of sequencing studies. Genome-wide association studies (GWAS) are instrumental in understanding the genetic architecture of complex traits. These studies typically identify numerous common genetic variants, each of small individual effect (1) that can be used in summation to make predictions. The idea that these common variants adequately represent the genetic architecture is supported by evolutionary theory, which suggests that many human traits (e.g., height, birth weight) are under stabilizing selection, which reduces the fitness impact of large-effect variants (2).

A recent hope for advancing personalized medicine is in the potential of polygenic risk scores to identify high-risk individuals who may benefit from early interventions, such as adoption of specific lifestyle regimes or increased screening (3–6). However, this strategy will be underpowered when applied to traits or diseases with reduced PRS in the tail of the distribution due to reduced common polygenicity. To overcome this problem, the genetic aetiology underlying the tails of phenotype distributions must be investigated and better characterized (7). Polygenic prediction models use these common small-effect variants to generate polygenic risk scores (PRS) whose performance making out-of-sample predictions is usually evaluated with aggregate estimates like *variance explained* (R^2). However, new evidence has suggested that this level of analysis may not be adequate because the preformance of prediction models is not uniform across the trait distribution. This non-uniformity results because the contribution of small effect common variants relative to highly penetrant rare variants and the environment is not equal across the trait distribution (7, 8).

This negative correlation between effect size and minor allele frequency (MAF) reflects the interplay between long term selection against extreme trait values and the ability of mutant alleles with significant effects to escape this trend through chance or short term environmental changes. Recent or ongoing selection can leave distinct genetic signatures, such as an enrichment of rare variants in individuals in the tails of trait distributions during directional or stabilizing selection. The expectation that individuals with extreme trait values are more likely to carry rare large-effect variants has led to strategies that priorize sequencing of individuals in the tails of the trait distribution (9, 10). However, tail trait values can also result from extreme polygenic burden created by recombination and reassortment, and a more efficient approach would involve considering more lines of individuals to allow the selection of individuals with strongest evidence of rare variant architecture for sequencing.

Here we attempt to distinguish whether the tails of the distribution of different traits are primarily driven by common polygenicity or by a more "monogenic-type" architecture **NEED: monogenic literally means single-gene disorder**. We accomplish this by developing a statistical test that measures the fit of the polygenic model across the trait-PRS distribution and identifies non-polygencity in each tail and applying it to over a **NEED: thousand** UK Biobank traits (see ***Figure* ??**) **NEED: sentence is too long and "trait-PRS distribution" is inaccurate description**. Next we consider whether evidence of negative selection exists in a subset of uncorrelated traits **NEED: we use all 75 traits, these are not uncorrelated** and show that the trait tails under negative selection are more likely to be associated with **NEED: PRS regression to the mean which may imply existence of rare variants**. We further triangulate evidence for selective constraint leading to reduced contribution of common polygenicity in the tails, by testing for the associations between known measures of fitness and PRS regression to the mean in the tails. (11, 12).

Finally, we assess whether the predictions of a complex trait architecture derived from a conditional sibling model align with the observed diversity in the architecture of trait extremes inferred from population level analyses.

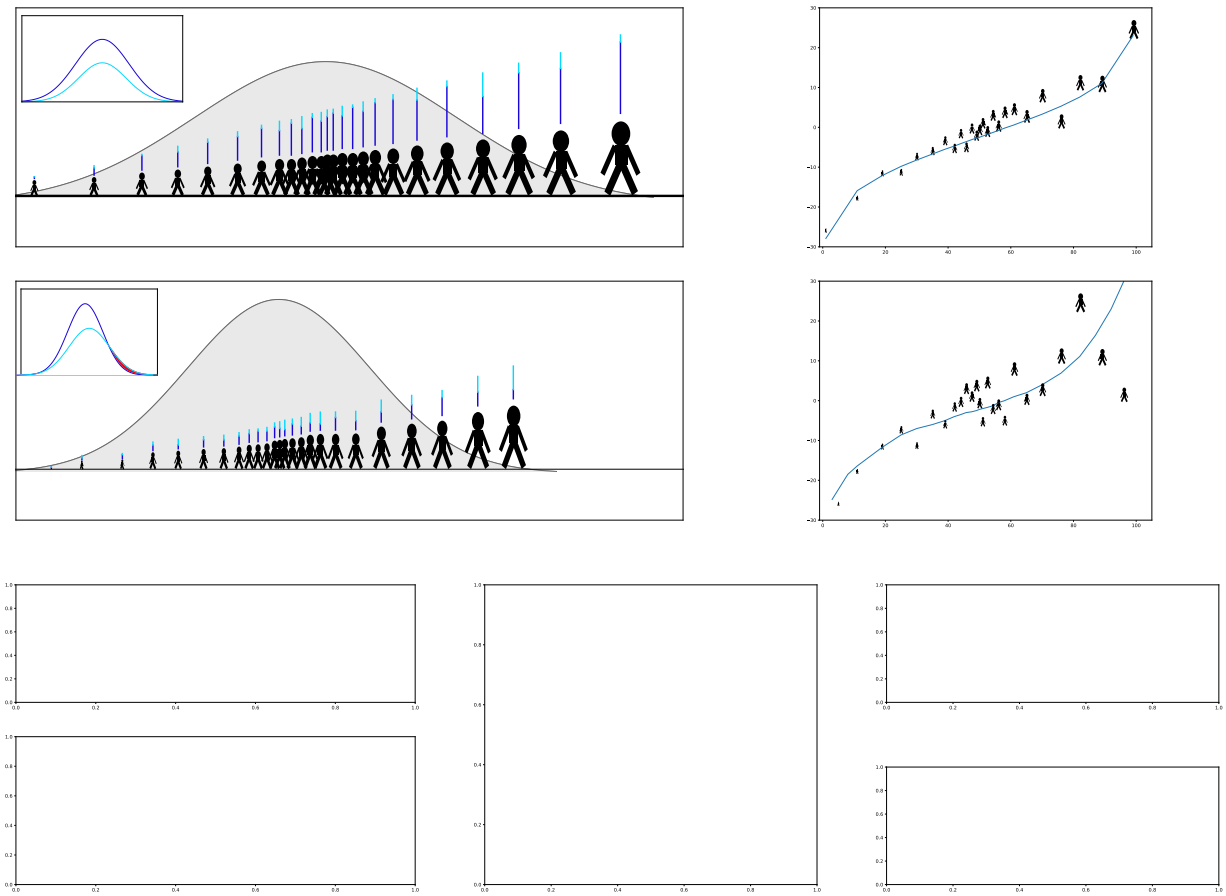


Fig. 1: Model Schematic and Simulated Results (d) Simulated Result: Distribution after stabilising selection. **(e)** Simulated Result: Trumpet plot before and after stabilising selection. **(f)** Simulated Result: Contribution of rare variants across the distribution under stabilising selection. **(g)** Simulated Result: Trait quantile vs mean quantile calculated using all variants (top) and only variants with MAF>1%.

Results

Motivated by **NEED: simulated results that demonstrating the relationship between selection, rare variants and POPout (see *Figure 1*)**, we selected UKB biobank traits for analyses from the following categories *Biomarkers*, *Physical Measures*, *Touchscreen Questionnaire* and *Cognitive Tests*. Continuous traits from these categories were standardised by regressing out the effects of age, sex, and known environmental and technical factors and then applying an inverse-normal transformation (see *Methods: Trait Normalisation*). Samples were then split in half, GWAS was carried out using Plink (6), and heritability and pairwise genetic correlation was estimated using LDscore (13). Applying quality controls for sample size, heritability, skew, and oligogenicity and removing strongly correlated traits resulted in the set of 74 traits (see *Methods: Trait QC*), used for the POPout associated analyses conducted across the large number of European ancestry samples in the UK Biobank and replicated three ways: across a subset of repeated measures European ancestry samples in the UK Biobank, acorss a subset of non-European ancestry samples in the UK Biobank, and, where applicable, across European ancestry samples in the All Of Us Biobank (see *Replication*).

The POPout Test

Conceptually, POPout occurs when when samples in the tails of the trait phenotype distribution have polygenic scores that significantly depart from expectation. To quantify this phenomenon we have developed the *POPout test* which regresses PRS-On-Phenotype to estimate the effect size, direction, and significance of departure from expected PRS for individuals in the lower and upper 1% of the trait distribution (see *Methods*).

Applying this test to the European ancestry samples in the UK-biobank reveals widespread departure from expectation across the 74 traits analysed. Correcting for multiple tests (5% false discovery threshold), 68 of 74 traits have significant POPout effects in one or both tails and that the vast majority (90%) of these effects are positive - i.e. reflect regression to the mean or unexpectedly mild risk scores - which is consistent with rare variants of large effects in individuals of the trait tails. ***Figure 2a*** illustrates the three broad types of tail effects by plotting trait quantile, in 1% increments, versus mean PRS in the quantile for four index traits:

- *Phosphate* - Selected because it has no observable $POPout$ Effect in either tail.
- *Haemoglobin Concentration* - selected for most significant POPout regression in lower tail only,
- *Red Blood Cell Distribution Width* - selected for most significant POPout regression in upper tail only,
- *Sitting Height* - selected for most significant POPout regression in both tails (minimum value).

Figure 2b enumerates the POPout tail effects for all traits analyzed, separated into three logical UKB categories, *Biomarkers*, *Physical Measures* and *Questionnaire/Cognitive*. Again positive POPout effects (regression to the mean) predominate, especially with respect to larger effects: 22 tails have positive POPout effects greater than 0.25 while none have negative POPout effects less than 0.25. Here we also observe that POPout regression is most common in *Biomarkers* category (79% of tails) and least common in *Questionnaire/Cognitive* category (43% of tails). Overall we observe a moderate relationship ($R = 0.23, Pv = 0.018$) between POPout regression and trait heritability, suggesting that POPout regression is not driven primarily by environmental effects.

Figure 2c illustrates the impact that tail $POPout$ has on the reliability of PRS model to predict phenotype in the upper tails. For both traits shown, *Haemoglobin Concentration* and *Red Blood Cell Distribution Width* the relative odds ratios are equal to expectation when "case" is defined very liberally - as being in lower or upper 25% of the distribution but underperform in the presence of $POPout$ when "case" is defined conservatively - being in the lower or upper 1% of the distribution. With respect to *Haemoglobin Concentration*, where lower tail $POPout$ is present, the relative odds ratio for the lowest PRS quantile (5%) when case is defined as being in the lower 1% is only 1.5, which is far below expectation and far below the upper tail analogue (top quantile (95% predicts case defined by upper 1%) which is 3.5. With respect to *Red Blood Cell Distribution Width*, where upper tail $POPout$ is present, the relative odds ratio for the largest PRS quantile (95%) when case is defined as being in the upper 1% is only 1.7, which is far below expectation and far below the lower tail analogue (bottom quantile (5% predicts case defined by lower 1%) which is 5.1. This shows that while the presence of $POPout$ in one or both tails does not preclude overall predictive power as defined using PRS r^2 it can cause the PRS to obtain all its predictive power by using the moderate range of the distribution, causing PRS models to describe the genetic architecture that underlies a range of moderate phenotypes that excludes the extremes where disease is more likely to manifest.

In figure ***Figure 2d*** the results of three flavours of $POPout$ replication performed on $POPout$ effect sizes is shown. In each replication dataset, matching traits with at least five thousand samples that passed our $POPout$ *regression QC* mentioned above were included and the Pearson correlation across the $POPout$ tail effect sizes was calculated.

Replication across the mean in a small samples of repeated measures (50 traits, $R = 0.71, Pvalue = 2.3 \times 10^{-16}$) in UKB samples of European ancestry demonstrates that $POPout$ effects are unlikely to be caused by random measurement errors that drive individuals to the tails of the distribution. Replication in UKB-samples of Non-European ancestry (54 traits, $R = 0.63, Pvalue = 3.7 \times 10^{-13}$) demonstrate that $POPout$ is not an ancestry specific phenomenon. Lastly, replication in the $AllofUs$ cohort (28 traits, $R = 0.59, Pvalue = 1.6 \times 10^{-6}$) demonstrate that $POPout$ is not a UKB specific phenomenon as a matched ancestry sample in a different country also has similar results. For more information on replication please see *supplement*. Interestingly, the correlation across ancestry (UKB European and non-European) samples is higher than the within ancestry across biobank correlation (UKB European and All of Us European) which suggests that despite accurate PRS prediction not being portable across ancestry that POPout regression is still portable across ancestry groups.

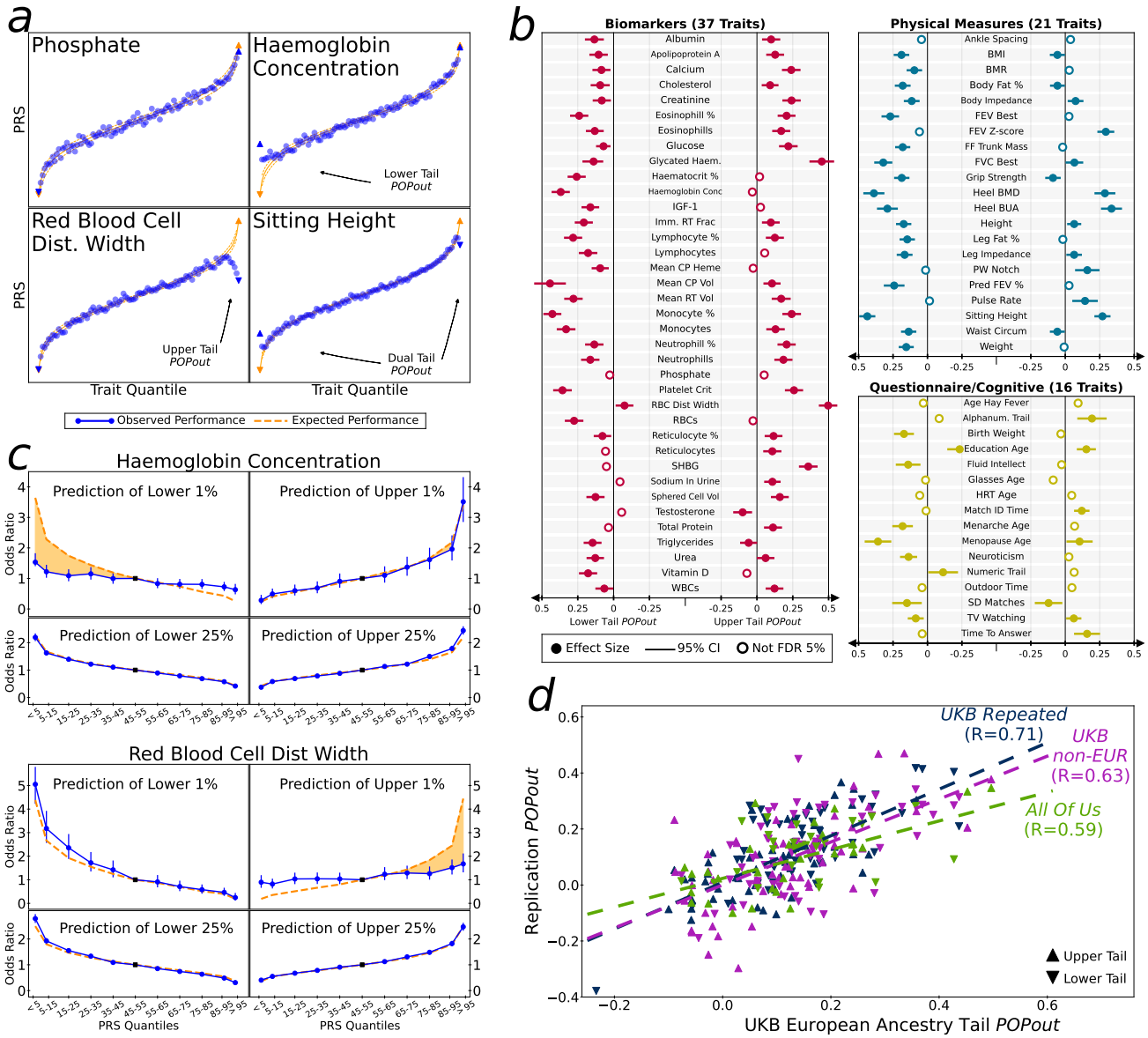


Fig. 2: **Tail POPout.** **(a)** PRS quantile (in 1% increments) plotted against mean trait value for four traits with different POP-out results (clockwise: No $POPout$ Effects, $POPout$ in Upper Tail, Lower Tail, Dual Tail). **(b)** POPout results for all 74 traits separated by category. **(c)** Case Control Accuracy Test **(d)** Replication

Replication in Siblings

Discordance between siblings in the tails of the trait distribution is indictive non-polygenic inheritance (14). Furthermore, excess trait concordance between siblings can be indicative of shared Mendelian effects (15). ***Figure 3***a illustartes the expected impact of polygenicity, de novo, and Mendelian rare variants within sibling pairs. Expected sibling pair concordance under polygenicity can be used to test for (1) excess discordance between siblings in tails of trait distribution, indicative of de novo mutations of large in one sibling and (2) excess concordance between siblings in tails of trait distribution, indicative of shared Mendelian effects (15). These tests are derived from the conditional sibling distribution under polygenicity, this distribution also allows heritability to be estimated (15).

Applying this model to the sibling data in the UKB to estimate h^2, we find a correlation between snp h2 and sibling h2 in ***Figure 3***b. As shown in ***Figure 3***c, we use this model to infer the presence of rare variant architecture in our index traits and find lower tail discordance in *Haemoglobin Concentration*, upper tail discordance in *Red Blood Cell Distribution Width* and dual ail discordance in *Sitting Height*, all of which match the result of the POPout analyses. We also find an overall significant relationship between sibling test pvalue and the POPout pvalue across all 74 traits ***Figure 3***d. Interestingly, this relationship varies across category, pearson correlation is 0.80 across the *Biomarker* group, 0.60 across the *Physical Measures* and insignificant the lower heritability *Questionnaire/Cognitive* category suggesting again that concordance between POPout effects and sibling discordance is not the result environmental causes alone.

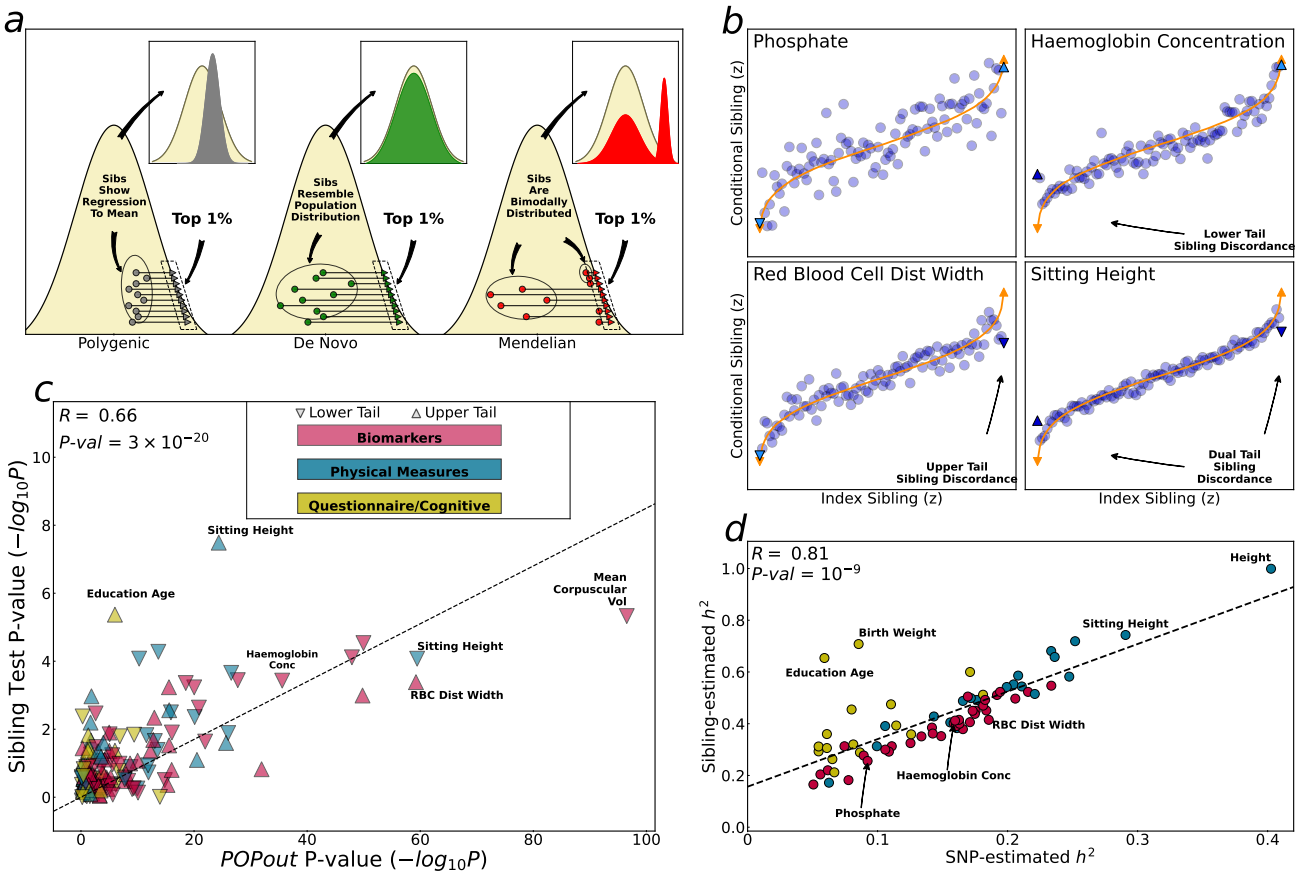


Fig. 3: $POPout$ **Replication in Siblings. (a)** Schematic of relationship of sibling trait values under polygenicity, de novo variants of large effect in one sibling and rare inherited variant of large effect, Mendelian. **(b)** LDscore regression SNP heritability plotted against heritability estimated from sibling pair trait similarity using European UKB samples. **(c)** Index sibling trait quantile (in 1% increments) plotted agaisnt mean trait value of conditional sibling in quantile for four traits with different types of PRS tail regression. **(d)** PRS-test P-values plotted against sibling de novo and Mendelian test meta analysis P-values on $-\log_{10}$ scale for upper and lower tails of all 74 traits. Correlation calculated using pearson correlation.

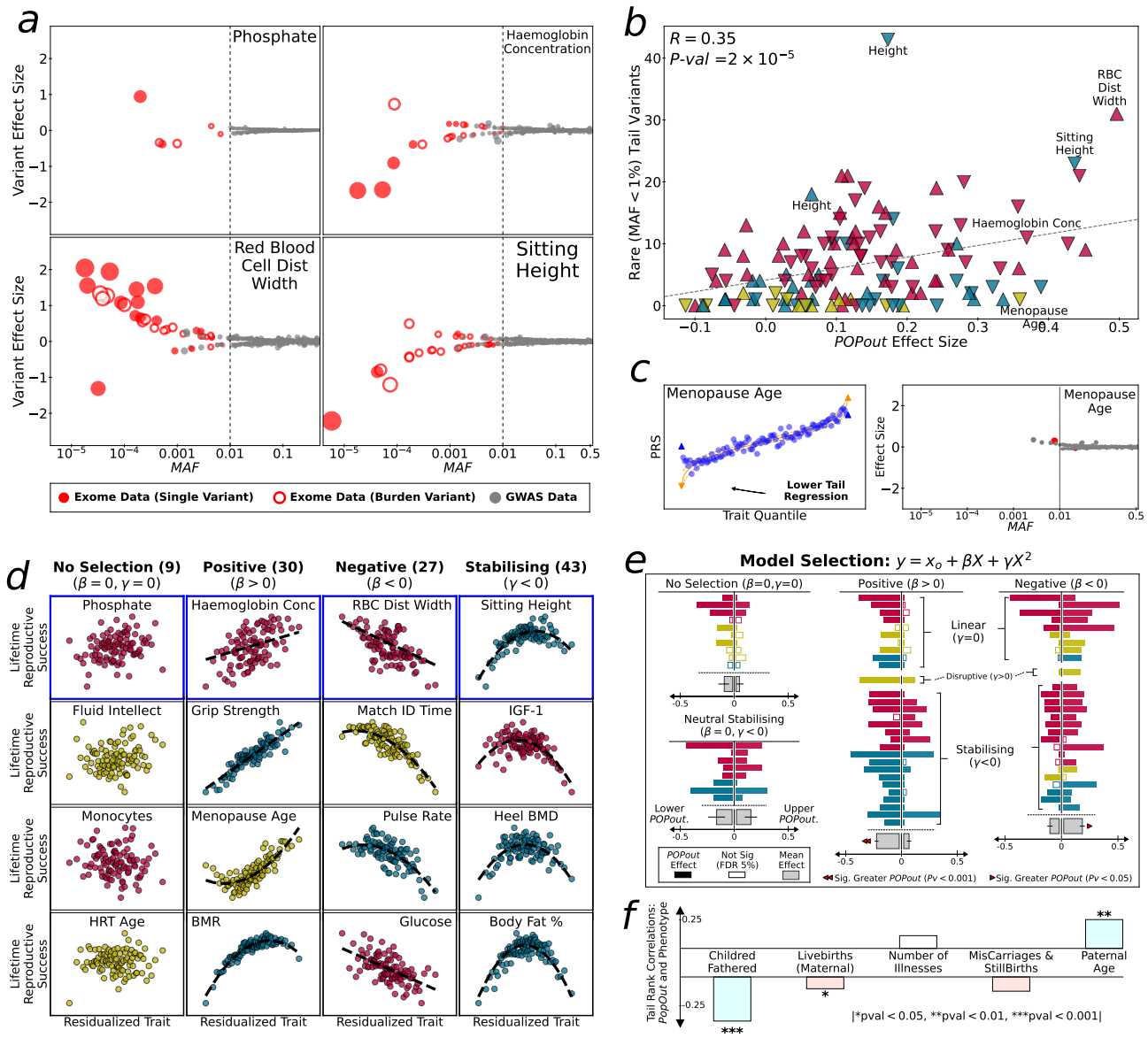


Fig. 4: **Rare Variant and Selection Analysis (a)** Trumpet plots (minor allele frequency v effect size) of genome-wide significant variants for traits shown in (a). **(b)** POP-out effect sizes plotted against the number of genome-wide significant variants with MAF<1% for 58 traits with overlapping exome sequence results. For upper tails positive variant effects are counted and for lower tails negative variant effects are counted. Correlation and significance calculated using pearson correlation. **(c)** Age at Menopause: PRS quantile vs. trait mean and trumpet plot. **(d)** Residualised trait value plotted against relative lifetime reproductive success for 16 example traits. Best model fit to the data, selected by BIC considering all models with and without linear and quadratic terms, are shown by dashed lines. Model selected by BIC used to classify selective pressure acting on the traits. **(e)** $POPout$ effect size shown by model selection category, with PRS-test effect size meta-analysis for each category. **(f)** Spearman rank correlation between four proxy measures of fitness and PRS-test effect size. For each fitness measure correlation was calculated between the mean of the fitness measure in individuals in the upper and lower 1% tails and PRS-test effect size for the corresponding tail across 74 traits in UKB passing QC.

The genetic mechanism proposed to explain $POPout$ involves the confluence of genetic selection away from one or both tails, an enrichment of rare variants in those tails, and PRS methods that typically restrict to common variants (MAF $> 1\%$) that suffers in such tails. Indeed, the exclusion of rare variants from PRS has already been shown to reduce PRS performance (16). ***Figure 4***a shows the relationship between MAF and effect sizes from significant associations derived from GWAS (17) and exome analyses (18) of UKB for our four index traits. For *Haemoglobin Concentration* which exhibits lower tail regression we see a greater number of rare variants in the lower tail and for *Red Blood Cell Distribution Width* which exhibits upper tail regression we see a greater number of rare variants in the upper tail. Sitting height, which exhibits dual tail regression, has a number of rare variants in both tails, however, there are more rare variants in the lower tail where the POPout effect is greatest. For *Phosphate* which does not exhibit POPout in either tail there are much fewer rare variants of large effect. These results are consistent with the hypothesis that rare variants of large effect that not accounted for in the

common-variant PRS are responsible for tail POPout effects.

Indeed this relationship holds up across all traits analysed. In In ***Figure 4***b we observe a small but significant relationship between the magnitude of *POPOut* regression in the tail and the number of rare variants in the tail across all trait tails analysed. However, we note that while statistically significant, there are multiple traits with significant *POPout* scores for which almost no rare variants of substantially large effect have been identified (see ***Figure 4***c). This suggests that strong selection may result in private variants that cannot be identified using population based sequencing.

The Impact of Selection

Using adjusted lifetime reproductive success (2, 19) as a proxy for selection we classified the selective pressure acting on each trait (see *Methods* - LRS) by fitting it an optimal two parameter model ($Y_{LRS} = x_o + \beta x + \gamma X^2$) as determined by BIC. Shown in ***Figure 4***a, most traits (65 of 74%) have some relationship (ie. $\beta \neq 0$ or $\gamma \neq 0$) between between adjusted lifetime success and trait quantile and the majority of them (43 of 65) two thirds of traits (43 of 65) were classified as being under some form of stabilising ($\gamma < 0$) selection, which is characterized by lower fecundity in the both tails of the trait distribution. This may be the result of general fitness favoring the trait mean or the result of sex specific selection that averages to produce this relationship (2). Only two traits were classified as under disruptive selection ($\gamma > 0$) which is characterised by greater fecundity in both trait tails.

Irrespective of their gamma parameter, 57 trait models included a nonzero directional beta parameter. In ***Figure 4***b we see that traits under positive selection have significantly more POPout regression in the lower tail and that traits under negative selection have signficicantly more POPout regression in the positive tail - which is in line with theoretical and simulated predictions that suggest that negative selection drives an enrichment of rare variants in the tail which reducing polygenicity and driveout POPout regression.

Tail Fitness

In addition to model selection on adjusted lifetime reproductive sucess we analysed the relationship between five ranked fitness related measures associated with *de novo* etiology and ranked POPout regression across all 148 tails analysed. Shown in ***Figure 4***c, we find that while POPout regression is associated with less less offspring in both sexes, that this effect may be stronger in males. Additionally we find that while neither the number of illnesses or miscarriages and stillbirths are significantly associated with POPout regression that paternal age (20) further suggesting that private variants may be driving some of the POPout signal.

Discussion

In this study, we propose the use of PRS Regression-to-the-Mean (RM) Test as a way to identify traits that are enriched for rare or de novo – more "monogenic-type" effects (MTE) in extreme. We applied this method to a list of quantitative traits available in UKB. In addition to genotype data, we also incorporated the rich source of UKB information including the sibling data (Sibling RM Test), and paternal age (Paternal Age Test) and number of children (Fecundity Test) into our analyses. We found directional consistency in most of the tests performed, providing converging evidence for the feasibility of using PRS as a strategy to identify traits with potential MTE. In our analyses, the PRS RM test can be used to identify both the "polygenic" and "MTE" event, we hypothesize that for a highly polygenic trait, samples with extreme trait values (ETV) are expected to have extreme PRS correspondingly. Deviation from the expectation may suggest samples at the extreme tails are enriched for MTE that are under selection. Our results show significant correlation in the ranking traits from least to most evidence of common polygenicity in the tails of their distributions between the PRS RM test and Fecundity test. Furthermore, the results of the PRS RM test are supported by the Sibling RM test of which we use the sibling phenotype data to investigate the common polygenicity in different traits. Sib-pairs are expected to have similar trait value under the influence of common polygenicity. On the other hand, if the ETV in sib A is contributed by de novo mutations, we expect sib B to be discordant phenotypically. An alternative explanation for discordant phenotype in sib-pairs could be due to the non-shared environment factors with large effect.

The striking concordance between these measures underscores the need to consider the role of rare variants in the analysis of trait tails. The PRS and sibling analyses presented here offer valuable tools for the targeted selection of traits and samples likely to harbor de novo or rare deleterious alleles and represent an approach that has the potential to significantly enhance the effectiveness and design of sequencing studies.

One major limitation of these tests is that they are highly sensitive to measurement errors. Measurement errors would result in the same PRS/ sibling departure at the extreme ends, because the PRS is no longer reflective by one's phenotypic measurement. In addition, both the PRS RM and sibling RM Test cannot distinguish between large effect mediated by environmental factors and MTE. For example, if a person with extremely low levels of LDL but normal PRS can be caused by malnutrition. Therefore, collection of the associated environmental factors and the corresponding adjustment becomes vital to alleviate this problem when applying these tests. Moreover, these tests are only applicable to quantitative traits, more work is needed to develop for testing binary outcome. In this study, we identified traits with evidence to be influenced by MTE at the trait tail(s), however, we did not identify or confirm the corresponding de novo mutation. Despite the lack of parent-offspring trios which is essential for the identification of de novo mutations, it is technically challenging to define a list of candidate genes and variants for every trait we investigated. Further investigation is warranted to validate this finding. In conclusion, we propose a novel way of using polygenic risk score to look for evidence of de novo/ rare mutations. This can be used as a method to identify traits that are more likely influenced by de novo mutations, and select samples who are more likely carry de novo mutations, potentially maximizing the power of a sequencing study. This method can be applied to any traits with continuous measure such as age of onset, disease severity and rate of disease progression.

Methods

Data Processing

UK Biobank Genetic Data

The UK Biobank (UKB) is a prospective cohort study of approximately 500,000 participants recruited across the United Kingdom from 2006 to 2010 (21). Phenotype data of anthropometric, biological and lifestyle measures was collected at baseline and follow-up surveys, with further linkage to health and disease record data. The genetic dataset consists of 488,377 samples genotyped at 805,426 SNPs. To determine population ancestries, 4-means clustering analysis was conducted on the first two principal components (PCs) of the genotype data. Samples were then projected onto 1000 Genomes PCs to determine ancestry, resulting in 461,931 European, 11,074 South Asian, 7,935 African, 2,585 West Asian, 2,550 East Asian, 1,619 and admixed samples.

Subsequent to clustering, standard quality control (QC) procedures were applied independently to each ancestry cluster. SNPs with a minor allele frequency < 0.01, genotype missingness > 0.02, or Hardy-Weinberg equilibrium test P-value $< 10^{-8}$ were excluded. Samples exhibiting high levels of missingness or heterozygosity, inconsistencies in genetic-inferred and self-reported sex, or displaying aneuploidy of the sex chromosomes were removed in accordance with recommendations from the UKB data processing team (22). For the analyses on the general population, a greedy algorithm was employed to remove related individuals that maximises sample retention while removing all third-degree relatives (kinship coefficient > 0.044) (23). After this QC, 411,949 unrelated individuals remained of which 387,235 were of European ancestry. Included among the European ancestry cluster were 18,340 individuals which repeated measures that were set aside along with the 24,476 non-European individuals for replication analyses. All told, this resulted in up to 369,133 unrelated European ancestry individuals for primary analyses.

For sibling analyses, sibling pairs were selected by first identifying first-degree relatives, defined as those with kinship coefficient within the range 0.177 and 0.354 (expected value for siblings is 0.25) to account for the expected variation in within-pair similarity (24). Only individuals of European ancestry were included in the sibling analyses. The proportion of SNPs with zero identity-by-state (IBS0) was used to distinguish sibling pairs from parent-offspring pairs, since parent–offspring pairs have IBS0≈0 across the autosomes. It has been shown in the UKB that there is good separation between pairs of first degree relatives at IBS0=0.00112 (24) so sibling pairs were selected as those exceeding this threshold, giving a total of 17,289 sibling pairs of European ancestry for analysis.

Quantitative Trait QC

Trait Selection

After genetic QC, we initially considered all traits from the categories *Biomarkers* (453 traits), *Physical Measures* (271 traits), the *Touchscreen Questionnaire* (396 traits) and *Cognitive Function* (134 traits) to provide a broad group of traits likely to have polygenic etiology. We then removed 34,698 cancer cases, 1,311 individuals on insulin, and 109 pregnant women from all traits. We also removed 64,043 individuals on statins and other cholesterol lowering drugs from the *Blood count* and *Blood Biochemistry* subcategories and removed 29,395 individuals on heart rate medication from the *Arterial stiffness* and *Blood pressure* subcategories. Next, to ensure that extreme outliers do not impact our results, we removed all samples from traits more than six standard deviations from the mean. After these removals, we defined quantitative traits as those which contained at least ten distinct values and whose top two modal values comprise less than half the samples. Finally, we removed all traits with fewer than $100,000$ samples remaining in our primary dataset of unrelated individuals of European ancestry, resulting in a set of 180 quantitive traits which were then normalised as described below.

Trait Normalisation

To minimise the effect that environmental risk factors have on phenotype, trait values were normalised within each ancestry group (European and Non-European) by applying a linear regression model to each trait, with covariates for age, age^2, age^3, sex, menopause status, type-II diabetes status, coronary artery disease status, alcohol intake, smoking (categorical - Field ID: 20116 and continuous - Field ID: 1239), batch, centre, and the first 40 genetic principal components. At this stage, traits whose residuals had absolute skew > 2 (25) in the larger European ancestry group were removed to avoid measurement bias and remaining trait values were standardized using a rank inverse normal transform on the residuals to provide standard normal distributions for all initial analyses.

Further Trait QC

To carry out our initial trait analyses, European ancestry UKB samples were split randomly into 50% base and 50% target data sets. Then, for each trait, GWAS was carried out on the standardised trait values in the base data using PLINK-1.9 (6, 26) and heritability and pairwise genetic correlation were estimated using LDscore (13). At this stage, to ensure that our PRS analyses are well powered and reflect additive polygenicity we removed traits with heritability $h^2 < 5\%$ (27) or for which oligogenicity was suggested due to a single SNP explaining more than 2% of trait variance (28). This left 120 remaining quantitative traits from across all four categories.

To verify that the traits were meaningfully different from each other, we considered the pairwise genetic correlations calculated above and observed that more than half (64 of the 120) remaining traits had pairwise genetic correlation $r_g > 95\%$ with at least one remaining trait. Of these 64 traits, 22 had a single pairwise correlation above this threshold. For these 11 pairs of traits, the trait with higher heritability was selected. The remaining genetically non-unique traits came from two categories *Bone-densitometry of heel* and *Body composition by impedance*. Applying the same data maximisation algorithm used to select the maximum number of unrelated individuals (23) we were able to select seven (two bone-densitometry related and 5 body composition related) without any pairwise genetic duplication (eg $r_g > 95\%$). In total addition to the 56 traits without genetic duplicates and the 11 traits selected from duplicated pairs, this resulted in the set of 74 traits which were used for all downstream analyses.

Initial PRS Analyses

The surviving 74 traits included 37 in the *Biomarkers* category, 21 traits in *Physical Measures* category, and 16 that were in either the *Touchscreen Questionnaire* or *Cognitive Function* category. In the primary unrelated European dataset these traits had between of 50,109 and 167,857 (average 123,482) samples in the base and target datasets. For each trait, PRSice (29) was used to train an optimal p-value threshold on the base data using the *FastScore* option and then applied to the target dataset to produce target risk scores that were combined with the normalised phenotype values in the target to run the $POPout$ test described below.

Replication Dataset QC

European Ancestry Repeated Measures

As discussed above, after initial filtering, there were 18,340 European ancestry individuals with measurements repeated after a five year-followup period. For these individuals, both measurements were normalised alongside the other European ancestry UKB samples but held out of the base dataset. Then the residualised trait values from their two visits were averaged to provide a target subset of individuals for which random measurement error is less likely to produce phenotype values in the tail. Across the 74 traits there were between 2,461 and 15,475 (average: 10699.4) unrelated individuals for repeated measure replication. Risk scores for this subset of data were calculated directly from the SNP weights generated in the UKB base data above.

Non-European Ancestry Individuals

The 24,476 Non-European ancestry individuals were filtered as above and normalised is a separate run that included all ancestry samples and risk scores were again generated using (29) on the 74 primary traits that were analysed in our primary dataset. Across these traits there were between 569 and 22,513 (average: 14,871.4) unrelated individuals for cross ancestry replication. Again, risk scores for this subset of data were calculated directly from the SNP weights generated above.

All of Us Cohort

All of Us is a multi-ancestry biobank from the United States that make use of electronic health record (EHR) data from a much wider age range than the UKB (μ, σ= 55.8,17.1 vs 56.5,8.1). Traits matching those analysed in our primary UKB analysis were identified through an initial database search and were then interrogated manually to determine if trait descriptions were sufficiently similar (see *Supplemental Table*. All of Us provides ancestry categories, corresponding to definitions used within gnomAD (30), the Human Genome Diversity Project (31), and 1000 Genomes project (32). 133,581 European ancestry individuals were extracted for analysis.

Trait QC in All Of Us was carried out to approximately match what was done in the UKB wherever possible. Like above, inidividuals on statins were removed from the *Blood count* and *Blood Biochemistry* subcategories. Then the median measurement value for every trait (EHR data includes multiple measures) and the age corresponding to that measurement was extracted. For each trait, outliers greater than six standard deviations from the mean were removed and a linear regression model was applied with covariates for age, age^2, age^3, sex, and the first 40 genetic principal components and a rank inverse normal transform was applied to residuals to provide standardized phenotype distributions.

Genetic QC in All Of Us was carried out with standard quality control procedures. SNPs with a minor allele frequency < 0.01, genotype missingness > 0.02, or Hardy-Weinberg equilibrium test P-value $< 10^{-8}$ were excluded. Samples exhibiting high levels of missingness or heterozygosity, inconsistencies in genetic-inferred and self-reported sex, or displaying aneuploidy of the sex chromosomes were removed and the same greedy algorithm mentioned above was used to maximize sample retention while removing all third-degree relatives (kinship coefficient > 0.044) (23). Using the higher powered publicly available European ancestry GWAS summary stats generated from the entire UKB (17), SNP weights and PRS scores were calculated (29) using the same p-value threshold trained in the UKB European ancestry analysis described above.

Data Analysis

Simulating Stabilizing Selection using SLiM

Using SLiM v4.0 (1), we simulated a quantitative polygenic trait under a standard Wright-Fisher (WF) model of stabilising selection. First a population of N=10,000 diploid individuals of 1Mb was simulated to equilibrium (10N generations) with a constant mutation of 2.36×10^{-8} per base pair (bp) and recombination rate of 10^{-8} per bp under neutrallity. Mutations were assigned effect sizes simulated from a gamma distribution with shape parameter 0.186 and mean 0.01026 and subsequently multiplied by -1 with probability 0.5 to give a symmetrical effect size distribution. This distribution has been used previously to simulate genetic effects in humans and reflects a prior belief in a small number of large effects and many small effects (33). Individual trait values were determined by the sum of all genetic effects.

After neutral genetic diversity was achieved (10N generations), stabilising selection on the trait was introduced for 1N generations through a Gaussian fitness function

$$f_t(x) = 1 + \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \quad \textbf{(1)}$$

where x is the phenotype of an individual at generation t, μ and σ are the trait mean and standard deviation at the start of selection (i.e., $t = 10N$). Stabilizing selection was simulated for a further 1N generations after the 10N burn-in generations.

Simulation data was collected by logging, at each generation, the phenotypic mean and standard deviation of the population, statistics of mutations present in the populations, including allele frequencies and effect sizes, and the mutational profile of each individual. To investigate the contributopn of low-frequency variants and mimic a PRS based on common variants, a second genetic value was calculate using only variants with $MAF > 1\%$.

POPout Analyses

The POPout Test

The POPout test first applies a regression of PRS on trait values where both the PRS and trait are standardised to have standard deviation of 1. For each trait, the residuals from the regression are split into quantiles according to the trait distribution and two-tailed t-tests are applied to test whether the residuals are significantly different from zero. The tests were applied to individuals in the top and bottom 1% of the trait distribution. The effect size in each tail is defined such that a positive effect is indicative of regression to the mean in both the upper and lower tails as follows

$$\frac{1}{n}\sum_{i}^{n}|f(y_i)| - |PRS_i|$$

where $f(y)$ is the predicted PRS from the regression given trait value y and n is the number of observations.

POPout regression QC

An additional QC was applied to ensure the robustness of the $POPout$ by testing the residuals of samples in the middle of the trait distribution for departure from the null of zero mean. Specifically, for each trait, two-sided t-tests for zero mean were applied to the residuals assigned in each of the middle 80 percentiles of the trait distribution. Traits were deemed to fail QC if the smallest P-value was significant at a Bonferroni-corrected level of 0.05 ($P \leq 0.05/80$).

Evaluating POPout Impact in Case/Control Paradigm

The predictive value of PRS in the tails of the distribution can be calculated by setting a case definition or tail threshold and calculating the odds-ratio under a case-control analysis. We considered two case definitions (the most extreme 1% of the trait distribution and the most extreme 25% of the trait distribution) and separated samples according to PRS quantiles that include the lowest 5%, between 5-15%, 15-25%, 25-35%, 45-55%,55-65%, 65-75%, 75-85%, 85-95% and the top 95%. Then for each trait threshold k (where case status is defined as being more extreme than k), the odds ratio and 95% confidence interval is calculated for each quantile by comparing it to the 45-55% quantile. Additionally, for each trait with PRS r^2, we simulated a trait where phenotype values were given by $T = rPRS + E$ where PRS has a standard normal distribution and the environmental effect E is drawn from a normal distribution with mean 0 and variance $(1 - r^2)$ to produce a simulated trait with a $\mathcal{N}(0,1)$ distribution and PRS r^2. Using the same number of target samples this simulation allowed us to calculate the expected relative odds-ratio in each quantile for comparison to our traits with differing levels of $POPout$ effect.

Replicating POPout Results

Bootstrap Analysis using European Ancestry Target Subsets

Our 74 QC-validated traits were required to have at least 100k total (50k target) samples. To explore whether our $POPout$ results replicate with smaller sample sizes, we repeatedly sampled subsets of between 1k and 50k target samples and performed our $POPout$ test on these subsets. For each value between 1k and 50k (counting by 5k) we performed 100 trials across each of the 148 trait tails. Implicitly assuming that the set of FDR 5% significant trait tails in our main analyses are "true positives" we calculated the empirical power for each subset size. We also calculated the linear correlation across $POPout$ effect sizes across the tails between each subset and our full dataset. This result informed our decision to require replication only in traits where at least 5,000 target samples are present - due to dramatically reduced power below this level. (see *Supplement Figure 1*).

Replication Across Datasets Motivated by the result above we replicated the $POPout$ effect in our three replication datasets, **(1)** UKB European ancestry repeated measures, **(2)** the UKB Non-European ancestry cohort, and **(3)** the European ancestry samples in All Of Us. In each dataset traits with at least five thousand samples that passed the $POPout$ *regression QC* mentioned above were included and the Pearson correlation across the $POPout$ tail effect sizes was calculated to quantify replication. Additionally, we also calculated the power or discovery rate by assuming that every tail effect significant at FDR 5% is a true positive.

Sibling concordance tests

The conditional distribution under polygenicity of an individual's trait value given their siblings trait value is predicted by theory (15)

$$p(s_2 \mid s_1) \sim \mathcal{N}\left(\frac{1}{2}s_1 h^2, 1 - \frac{h^4}{4}\right) \tag{2}$$

where s_1 and s_2 denote the index and conditional-sibling trait values respectively and h^2 is the trait heritability. Rare large-effect variants can lead to deviations from this distribution, particularly in the tails. Specifically, rare large-effect variants within families, *Mendelian effects*, will result in excess concordance between siblings when they both inherit the effect variant. Conversely, a large effect *de novo* mutation in one sibling will result in excess discordance. The sibling test for *Mendelian* effects applies a binomial test for a greater than expected number of sibling pairs both lying in a defined quantile of the tail of the trait distribution by calculating the following z-statistic

$$z = \frac{r - n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}}$$

where n is the number of index siblings in the quantile q under test, r is the number of siblings pairs both in q and π_0 is the expected proportion of sibling-pairs in the quantile under polygenicity

$$\pi_0 = \frac{1}{n}\sum_{i \in A_q}\left(1 - \Phi\left(\frac{\Phi^{-1}(q) - \frac{s_1^{(i)}h^2}{2}}{\sqrt{1-\frac{h^4}{4}}}\right)\right)$$

where A_q is the set of index siblings in trait quantile q, Φ is the normal normal cumulative density function, Φ^{-1} is its inverse and n is the number of sibling pairs.

The sibling test for *de novo* effects constructs a z-statistic for the distance between index siblings in the tail of the trait distribution and their conditional siblings as follows

$$z = \frac{\sum_{i \in A_q} s_2^{(i)} - \frac{1}{2}s_1^{(i)}h^2}{\sqrt{n(1-\frac{h^4}{4})}}$$

For index siblings in the lower and upper tails the alternative hypotheses of $H_1 : z > 0$ and $H_1 : z < 0$ are tested respectively. For further details see Souaiaia et al (15). Heritability h^2 required for the *Mendelian* and *de novo* tests is estimated from the sibling pairs by taking the maximum likelihood of h^2 under the null hypotheses of both tests

$$\log \mathcal{L} \propto -n \log\left(1 - \frac{h^4}{4}\right) - \frac{1}{1-\frac{h^4}{4}}\sum_{i=1}^{n}\left(s_2^{(i)} - \frac{s_1^{(i)}h^2}{2}\right)^2$$

The sibling meta P-value was calculated by combining P-values from the *Mendelian* and *de novo* tests using Fisher's method to give a χ^2 statistic with 4 degrees of freedom,

$$\chi_4^2 = -2(\log(p_{Mendelian}) + \log(p_{denovo}))$$

Rare Variant and Selection Analyses

POPout Rare Variant Associations

Backman et al (18) provides exome sequencing analysis across 3994 UKB traits including 65 of the 74 traits analysed in our target set. For each of these 65 traits, in each tail (positive and negative effect), the number of genes with at least one significant (P$\leq 2.18 \times 10^{-11}$) variant (deleterious, missense, predicted loss of function) or a genome-wide significant burden association are counted. Across the 130 trait tails, the pearson correlation between the number of exome variants and the $POPout$ effect size is then calculated.

Trumpet plots

Trumpet plots show the effect sizes and MAF of independent genome-wide significant variants. Common variants (MAF>1%) were selected as the most significant variant (P$\leq 5 \times 10^{-8}$) in 1Mb windows from publicly available GWAS summary statistics generated using UKB samples of European ancestry (17). Exome sequencing associations from the analysis of UKB European ancestry samples were downloaded (18). Results are plotted for the most significant deleterious, missense or predicted loss of function variants in each gene (P$\leq 2.18 \times 10^{-11}$, one association per gene). Genome-wide significant burden associations are shown where no single variant within the gene is genome-wide significant.

Relative Lifetime Reproductive Success

The number of offspring is recorded in the UKB for both females and males, fields *2734* and *2405* respectively. Following the methodology of Beauchamp (19), samples were split into four birth cohorts (1934-42, 1943-48, 1949-55, 1956-65). Individual relative lifetime reproductive success, Y_{rLRS}, was then calculated by dividing the number of offspring of each individual by the cohort mean. Following Sanjak et al (2), the presence and direction of selective pressure acting on traits was determined by fitting the following regression model:

$$Y_{rLRS} = \beta_o + \beta z + \gamma z^2 \quad \textbf{(3)}$$

where z denotes the residualised and normalised trait values and β and γ are linear and quadratic regression parameters.

Bayesian information criteria (BIC) was used to select the best fitting model for each trait from which the selective pressure acting on trait was inferred. If the selected model did not include a linear or quadratic parameter, no selection was inferred, here rare variant enrichment is not expected in either tail. If the selected model included only the linear parameter β, negative selection was inferred if $\beta < 0$ and rare variant enrichment is expected in the upper tail, while positive selection was inferred if $\beta > 0$ and rare variant enrichment is expected in the lower tail. If the selected model included only the quadratic parameter γ, with $\gamma < 0$, stabilising selection is inferred and the mean trait value in the population is at its optimum and rare variant enrichment is expected in the both tails. Disruptive selection is implied by $\gamma > 0$, in this scenario rare variant enrichment is not expected in either tail. Where the selected model included both non-zero linear and quadratic parameters then the β parameter is used to determine our expectation where $\beta < 0$ and $\beta > 0$ suggests greater relative rare variant enrichment in the upper and lower tails respectively.

Associating POPout Effect and LRS Models

Traits are separated into four classes. The first two of the four classes are non-directional, **(1)** no selection ($\beta = 0, \gamma >= 0$) where no rare variant enrichment is expected and **(2)** neutral stabilising selection ($\beta = 0, \gamma < 0$) where an equal amount of rare variant enrichment is expected each tail. The third and fourth class are directional, and include **(3)** positive selection ($\beta > 0$) where more rare variant enrichment is expected in the lower tail and **(4)** negative selection ($\beta > 0$) where more rare variant enrichment is expected int he upper tail. For each class the mean $POPout$ effect is calculated in each tail and the mean values are compared using a T-Test.

Association between POPout tests and proxy measures of fitness

Five proxy measures of fitness were considered, **(1)** Male and **(2)** female fecundity (Trait-Ids: 2405 and 2734), **(3)** self-reported non-cancer illnesses (Trait-Id: 2002), **(4)** sum of miscarriages and still-births (Trait-Ids: 39290 and 3839), and **(5)** paternal age (calculated as the difference between fathers age (Trait-Id 2946) and participant age (Trait-Id 21022). Association between these measures and $POPout$ effect was tested as follows. First, the mean value of the five proxies of selection was calculated separately in the upper and lower 1% of the trait distribution. Then, using Spearman's rank correlation, the association between mean tail values of each proxy and $POPout$ effect size across the tails was calculated.

1. William S Bush and Jason H Moore. Chapter 11: Genome-wide association studies. *PLoS computational biology*, 8(12):e1002822, 2012.
2. Jaleal S Sanjak, Julia Sidorenko, Matthew R Robinson, Kevin R Thornton, and Peter M Visscher. Evidence of directional and stabilizing selection in contemporary humans. *Proceedings of the National Academy of Sciences*, 115(1):151–156, 2018.
3. Alicia R Martin, Christopher R Gignoux, Raymond K Walters, Genevieve L Wojcik, Benjamin M Neale, Simon Gravel, Mark J Daly, Carlos D Bustamante, and Eimear E Kenny. Human demographic history impacts genetic risk prediction across diverse populations. *The American Journal of Human Genetics*, 100(4):635–649, 2017.
4. Brendan K Bulik-Sullivan, Po-Ru Loh, Hilary K Finucane, Stephan Ripke, Jian Yang, Schizophrenia Working Group of the Psychiatric Genomics Consortium, Nick Patterson, Mark J Daly, Alkes L Price, and Benjamin M Neale. Ld score regression distinguishes confounding from polygenicity in genome-wide association studies. *Nature genetics*, 47(3):291–295, 2015.
5. Frank Dudbridge. Power and predictive accuracy of polygenic risk scores. *PLoS genetics*, 9(3):e1003348, 2013.
6. Christopher C Chang, Carson C Chow, Laurent CAM Tellier, Shashaank Vattikuti, Shaun M Purcell, and James J Lee. Second-generation plink: rising to the challenge of larger and richer datasets. *Gigascience*, 4(1):s13742–015, 2015.
7. Patrick J Short, Jeremy F McRae, Giuseppe Gallone, Alejandro Sifrim, Hyejung Won, Daniel H Geschwind, Caroline F Wright, Helen V Firth, David R FitzPatrick, Jeffrey C Barrett, et al. De novo mutations in regulatory elements in neurodevelopmental disorders. *Nature*, 555(7698):611–616, 2018.
8. Brian J O'Roak, Pelagia Deriziotis, Choli Lee, Laura Vives, Jerrod J Schwartz, Santhosh Girirajan, Emre Karakoc, Alexandra P MacKenzie, Sarah B Ng, Carl Baker, et al. Exome sequencing in sporadic autism spectrum disorders identifies severe de novo mutations. *Nature genetics*, 43(6):585–589, 2011.
9. Dalin Li, Juan Pablo Lewinger, William J Gauderman, Cassandra Elizabeth Murcray, and David Conti. Using extreme phenotype sampling to identify the rare causal variants of quantitative traits in association studies. *Genetic epidemiology*, 35(8):790–799, 2011.
10. Zuoheng Wang, Xiangtao Liu, Bao-Zhu Yang, and Joel Gelernter. The role and challenges of exome sequencing in studies of human diseases. *Frontiers in genetics*, 4:160, 2013.
11. Cathie Sudlow, John Gallacher, Naomi Allen, Valerie Beral, Paul Burton, John Danesh, Paul Downey, Paul Elliott, Jane Green, Martin Landray, et al. Uk biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. *PLoS medicine*, 12(3):e1001779, 2015.
12. Ani Manichaikul, Josyf C Mychaleckyj, Stephen S Rich, Kathy Daly, Michèle Sale, and Wei-Min Chen. Robust relationship inference in genome-wide association studies. *Bioinformatics*, 26 (22):2867–2873, 2010.
13. Guiyan Ni, Gerhard Moser, Stephan Ripke, Benjamin M Neale, Aiden Corvin, James TR Walters, Kai-How Farh, Peter A Holmans, Phil Lee, Brendan Bulik-Sullivan, et al. Estimation of genetic correlation via linkage disequilibrium score regression and genomic restricted maximum likelihood. *The American Journal of Human Genetics*, 102(6):1185–1194, 2018.
14. Abraham Reichenberg, Martin Cederlöf, Andrew McMillan, Maciej Trzaskowski, Ori Kapra, Eyal Fruchter, Karen Ginat, Michael Davidson, Mark Weiser, Henrik Larsson, et al. Discontinuity in the genetic and environmental causes of the intellectual disability spectrum. *Proceedings of the National Academy of sciences*, 113(4):1098–1103, 2016.
15. Tade Souaiaia, Hei Man Wu, Clive Hoggart, and Paul O'Reilly. Sibling similarity can reveal key insights into genetic architecture. *eLife*, feb 2023. doi: 10.7554/elife.87522.1.
16. Zhe Wang, Shing Wan Choi, Nathalie Chami, Eric Boerwinkle, Myriam Fornage, Susan Redline, Joshua C Bis, Jennifer A Brody, Bruce M Psaty, Wonji Kim, et al. The value of rare genetic variation in the prediction of common obesity in european ancestry populations. *Frontiers in Endocrinology*, 13:863893, 2022.
17. Benjamin Neale. Neale lab data: http://www.nealelab.is/uk-biobank/, 2018.
18. Joshua D Backman, Alexander H Li, Anthony Marcketta, Dylan Sun, Joelle Mbatchou, Michael D Kessler, Christian Benner, Daren Liu, Adam E Locke, Suganthi Balasubramanian, et al. Exome sequencing and analysis of 454,787 uk biobank participants. *Nature*, 599(7886):628–634, 2021.
19. Jonathan P Beauchamp. Genetic evidence for natural selection in humans in the contemporary united states. *Proceedings of the National Academy of Sciences*, 113(28):7774–7779, 2016.
20. Augustine Kong, Michael L Frigge, Gisli Masson, Soren Besenbacher, Patrick Sulem, Gisli Magnusson, Sigurjon A Gudjonsson, Asgeir Sigurdsson, Aslaug Jonasdottir, Adalbjorg Jonasdottir, et al. Rate of de novo mutations and the importance of father's age to disease risk. *Nature*, 488(7412):471–475, 2012.
21. Cathie Sudlow, John Gallacher, Naomi Allen, Valerie Beral, Paul Burton, John Danesh, Paul Downey, Paul Elliott, Jane Green, Martin Landray, et al. Uk biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. *PLoS medicine*, 12(3):e1001779, 2015.
22. Information for Researchers. Ukb genotyping and quality control (v.1.2): https://biobank.ctsu.ox.ac.uk/crystal/crystal/docs/genotyping_qc.pdf, 2015.
23. Shing Wan Choi, Timothy Shin Heng Mak, Clive J Hoggart, and Paul F O'Reilly. Erasor: a software tool to eliminate inflation caused by sample overlap in polygenic score analyses. *GigaScience*, 12:giad043, 2023.
24. Rosa Cheesman, Jonathan Coleman, Christopher Rayner, KL Purves, Genevieve Morneau-Vaillancourt, Kylie Glanville, Shing W Choi, Gerome Breen, and Thalia C Eley. Familial influences on neuroticism and education in the uk biobank. *Behavior genetics*, 50(2):84–93, 2020.
25. Paul Mallery and Darren George. *SPSS for windows step by step*. Allyn & Bacon, Inc., 2000.
26. Shaun Purcell, Benjamin Neale, Kathe Todd-Brown, Lori Thomas, Manuel AR Ferreira, David Bender, Julian Maller, Pamela Sklar, Paul IW De Bakker, Mark J Daly, et al. Plink: a tool set for whole-genome association and population-based linkage analyses. *The American journal of human genetics*, 81(3):559–575, 2007.
27. Shing Wan Choi, Timothy Shin-Heng Mak, and Paul F O'Reilly. Tutorial: a guide to performing polygenic risk score analyses. *Nature protocols*, 15(9):2759–2772, 2020.
28. Arthur Korte, Bjarni J Vilhjálmsson, Vincent Segura, Alexander Platt, Quan Long, and Magnus Nordborg. A mixed-model approach for genome-wide association studies of correlated traits in structured populations. *Nature genetics*, 44(9):1066–1071, 2012.
29. Shing Wan Choi and Paul F O'Reilly. Prsice-2: Polygenic risk score software for biobank-scale data. *Gigascience*, 8(7):giz082, 2019.
30. Konrad J Karczewski, Laurent C Francioli, Grace Tiao, Beryl B Cummings, Jessica Alföldi, Qingbo Wang, Ryan L Collins, Kristen M Laricchia, Andrea Ganna, Daniel P Birnbaum, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581(7809):434–443, 2020.
31. L Luca Cavalli-Sforza. The human genome diversity project: past, present and future. *Nature Reviews Genetics*, 6(4):333–340, 2005.
32. 1000 Genomes Project Consortium Corresponding author Durbin Richard M. rd@ sanger. ac. uk 1 b, Production group: Baylor College of Medicine Gibbs Richard A.(Principal Investigator) Deiros David 14 Metzker Mike 14 Muzny Donna 14 Reid Jeff 14 Wheeler David 14, BGI-Shenzhen Wang Jun (Principal Investigator) Li Jingxiang Jian Min 22 Li Guoqing 22 Li Ruiqiang 22 23 Liang Huiqing 22 Tian Geng 22 Wang Bo 22 Wang Jian 22 Wang Wei 22 Yang Huanming 22 Zhang Xiuqing 22 Zheng Huisong 22, Broad Institute of MIT, Harvard Lander Eric S.(Principal Investigator) 2 Altshuler David 2 3 4 Ambrogio Lauren 2 Bloom Toby 2 Cibulskis Kristian 2 Fennell Tim J. 2 Gabriel Stacey B.(Co-Chair) 2 Jaffe David B. 2 Shefler Erica 2 Sougnez Carrie L. 2, Illumina Bentley David R.(Principal Investigator) Gormley Niall 6 Humphray Sean 6 Kingsbury Zoya 6 Kokko-Gonzales Paula 6 Stone Jennifer 6, Life Technologies McKernan Kevin J.(Principal Investigator) 24 Costa Gina L. 24 Ichikawa Jeffry K. 24 Lee Clarence C. 24, Washington University in St Louis Mardis Elaine R.(Co-Chair)(Co-Principal Investigator) Wilson Richard K.(Co-Principal Investigator) Dooling David 17 Fulton Lucinda 17 Fulton Robert 17 Weinstock George 17, et al. A map of human genome variation from population-scale sequencing. *Nature*, 467 (7319):1061–1073, 2010.
33. Xin He, Stephan J Sanders, Li Liu, Silvia De Rubeis, Elaine T Lim, James S Sutcliffe, Gerard D Schellenberg, Richard A Gibbs, Mark J Daly, Joseph D Buxbaum, et al. Integrated model of de novo and inherited genetic variants yields greater power to identify risk genes. *PLoS genetics*, 9(8):e1003671, 2013.

Supplement

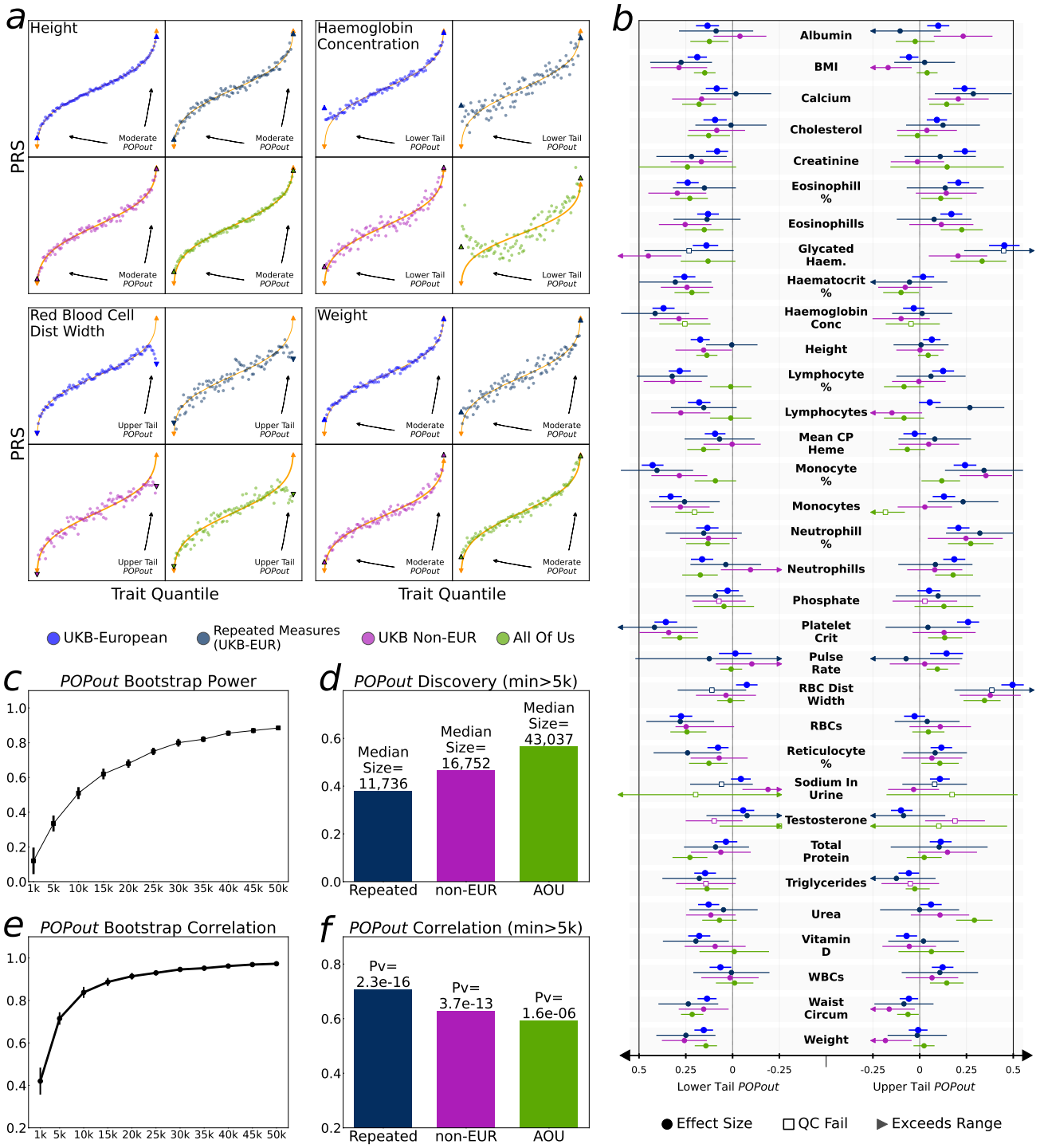


Fig. 5: **Extended Replication (a)** PRS quantile (in 1% increments) plotted agaisnt mean trait value in quantile for four traits with different types of PRS tail regression for discovery and replication cohorts. **(b)** PRS-test effect size in discovery and replication cohorts across 25 traits measured and passing PRS-test quality control in all cohorts. **(c-e)** PRS-test effect sizes estimated in UKB European ancestry discovery samples and replication cohort: **(c)** UKB European ancestry sample repeated measures, **(d)** UKB non-European ancestry samples and (e) AllofUs European ancestry. Triangles shaded relative to the replication sample size for the trait. Pearson correlation shown. samples. **(f)**....

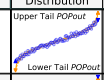
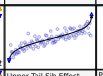
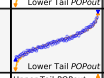
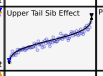
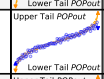
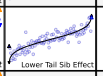
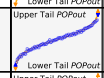
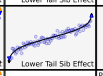
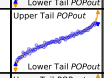
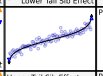
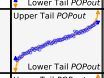
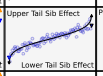
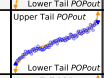
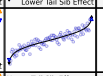
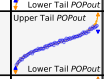
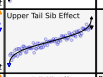
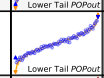
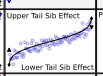
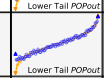
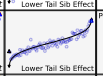
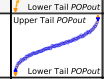
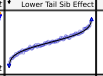
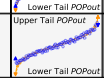
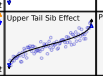
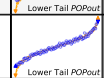
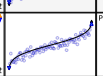
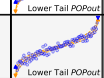
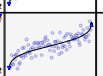
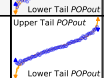
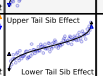
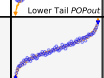
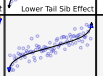
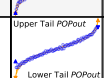
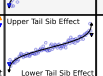
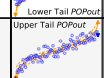
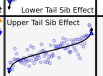
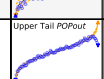
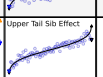
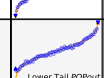
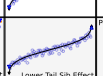
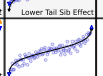
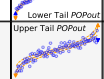
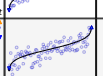
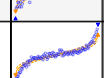
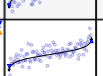
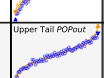
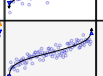
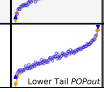
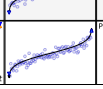
Trait Name	Overview	*POPout* Tests (Upper | Lower)	Sib Tests	Replication	Rare/Evolution	POP (Phenotype on PRS) Distribution	Conditional Sibling Distribution	Selection Inference Distribution
Category, Abbreviated Name (AOU Name)	[UKB ID] Target Size,Sib Pairs SNP-h^2,SIB-h^2,PRS-r^2	Pval., 5% FDR, Effect Size, 95% CI	Meta P-values	Repeated Measures | Non-EUR | All Of Us Target Size, QC, pvals, effect sizes (Lower,Upper)	Rare Variants (Lower, Upper), Model Params: ($Y \sim X_0 + \beta X + \gamma X^2$)			
Albumin	[ID: 30600] 122k, 9k, 10%, 29%, 3%	6.8e-06, True, 0.13, [0.08,0.19]	0.876	8.6k, PASS, (0.373, 0.335), (0.09, 0.1)	2, 8			
Biomarkers Albumin Albumin [M/v] in Serum or Plasma		5.0e-04, True, -0.1, [-0.15,-0.04]	0.413	15.9k, PASS, (0.574, 0.003), (-0.04, -0.23) 73.5k, PASS, (7.2e-04, 0.524), (0.12, 0.02)	-0.018, 0, -0.004			
Body Mass Index (bmi)	[ID: 21001] 167k, 17k, 17%, 49%, 5%	2.9e-14, True, 0.19, [0.14,0.24]	0.022	15.4k, PASS, (0.001, 0.745), (0.28, -0.03)	1, 0			
Physical Measures BMI BMI-mean		0.017, True, 0.06, [0.01,0.1]	0.148	22.3k, PASS, (1.8e-04, 0.008), (0.29, 0.17) 127k, PASS, (1.9e-13, 0.022), (0.21, -0.06)	0.004, 0.034, -0.025			
Calcium	[ID: 30680] 121k, 9k, 10%, 30%, 3%	0.003, True, 0.08, [0.03,0.14]	0.437	8.6k, PASS, (0.845, 0.006), (-0.02, -0.29)	4, 4			
Biomarkers Calcium Calcium [M/v] in Serum or Plasma		5.0e-16, True, -0.24, [-0.3,-0.18]	0.022	15.9k, PASS, (0.037, 0.012), (0.17, -0.21) 83.4k, FAIL, (7.8e-08, 4.0e-05), (0.18, -0.14)	-0.019, 0, -0.003			
Cholesterol	[ID: 30690] 133k, 11k, 19%, 52%, 11%	0.002, True, 0.09, [0.03,0.15]	0.517	10.7k, PASS, (0.929, 0.207), (0.01, -0.12)	7, 11			
Biomarkers Cholesterol Cholesterol [M/v] in Serum or Plasma		3.9e-04, True, -0.09, [-0.14,-0.04]	0.007	17.4k, PASS, (0.264, 0.63), (0.08, -0.04) 64.4k, PASS, (0.006, 0.198), (0.12, -0.05)	-0.018, 0, -0.005			
Creatinine	[ID: 30700] 133k, 11k, 17%, 45%, 6%	0.004, True, 0.08, [0.03,0.14]	0.158	10.6k, PASS, (0.021, 0.249), (0.22, -0.11)	19, 10			
Biomarkers Creatinine Creatinine [M/v] in Serum or Plasma		9.1e-17, True, -0.24, [-0.3,-0.19]	0.23	17.4k, PASS, (0.043, 0.866), (0.17, 0.01) 83.9k, PASS, (2.7e-04, 8.8e-23), (0.13, -0.35)	-0.012, 0.02, -0.01			
Eosinophill Percentage	[ID: 30210] 129k, 10k, 16%, 39%, 6%	8.3e-17, True, 0.24, [0.18,0.3]	0.021	11.7k, PASS, (0.074, 0.186), (0.15, -0.14)	2, 12			
Biomarkers Eosinophill % Eosinophils/100 leukocytes in Blood by AC		1.7e-13, True, -0.21, [-0.26,-0.15]	0.036	16.7k, PASS, (1.8e-04, 0.082), (0.3, -0.14) 61.3k, PASS, (2.5e-12, 0.13), (0.28, -0.06)	-0.013, 0.02, -0.01			
Eosinophill Count	[ID: 30150] 129k, 10k, 16%, 37%, 5%	2.5e-06, True, 0.13, [0.08,0.19]	0.102	11.7k, PASS, (0.129, 0.44), (0.14, -0.08)	4, 10			
Biomarkers Eosinophills Eosinophils [#/v] in Blood by AC		2.1e-09, True, -0.17, [-0.22,-0.11]	0.322	16.7k, PASS, (3.7e-04, 0.177), (0.25, -0.12) 60.5k, FAIL, (0.016, 0.991), (0.1, 0.0)	-0.015, -0.02, -0.007			
Glycated Haemoglobin	[ID: 30750] 127k, 10k, 18%, 49%, 8%	8.2e-11, True, 0.14, [0.08,0.21]	0.001	8.3k, FAIL, (0.053, 4.3e-12), (0.23, -0.45)	9, 19			
Biomarkers Glycated Haem. Hemoglobin A1c/Hemoglobin.total in Blood		1.6e-50, True, -0.45, [-0.53,-0.37]	0.165	15.4k, PASS, (9.6e-07, 0.009), (0.45, -0.2) 38.9k, PASS, (0.364, 8.9e-09), (0.05, -0.29)	-0.063, -0.011, 0			
Haematocrit Percentage	[ID: 30030] 130k, 10k, 14%, 38%, 4%	2.9e-19, True, 0.26, [0.2,0.31]	0.036	11.7k, PASS, (0.002, 0.591), (0.31, 0.05)	8, 9			
Biomarkers Haematocrit % Hematocrit [Volume Fraction] of Blood by AC		0.517, False, -0.02, [-0.07,0.04]	0.003	16.8k, PASS, (5.9e-04, 0.284), (0.24, 0.08) 76k, FAIL, (3.5e-12, 0.002), (0.26, 0.11)	-0.022, 0.008, 0			
Haemoglobin Concentration	[ID: 30020] 130k, 10k, 15%, 40%, 4%	2.6e-36, True, 0.37, [0.31,0.42]	0.155	11.7k, PASS, (1.2e-05, 0.869), (0.41, -0.01)	8, 11			
Biomarkers Haemoglobin Conc Hemoglobin [M/v] in Blood		0.243, False, 0.03, [-0.02,0.09]	0.002	16.8k, PASS, (2.9e-04, 0.195), (0.29, 0.1) 83.8k, FAIL, (1.1e-08, 3.8e-05), (0.19, 0.15)	0.003, 0.005, 0			
Standing Height	[ID: 50] 167k, 17k, 40%, 99%, 21%	1.3e-12, True, 0.17, [0.13,0.22]	0.796	15.5k, PASS, (0.95, 0.921), (0.0, -0.01)	18, 43			
Physical Measures Height Height-mean		0.022, True, -0.06, [-0.02,-0.11]	0.096	22.4k, PASS, (0.041, 0.976), (0.15, -0.0) 127k, PASS, (2.0e-15, 0.052), (0.22, -0.05)	-0.002, 0, -0.019			
Lymphocyte Percentage	[ID: 30180] 129k, 10k, 13%, 35%, 4%	1.1e-22, True, 0.28, [0.23,0.34]	0.043	11.7k, PASS, (8.3e-04, 0.514), (0.32, -0.06)	10, 7			
Biomarkers Lymphocyte % Lymphocytes [#/v] in Blood by AC		9.7e-06, True, -0.13, [-0.18,-0.07]	0.078	16.8k, PASS, (5.9e-05, 0.949), (0.32, 0.0) NA	-0.014, 0.032, -0.008			
Lymphocyte Count	[ID: 30120] 129k, 10k, 17%, 44%, 5%	9.0e-10, True, 0.18, [0.12,0.23]	0.177	11.7k, PASS, (0.081, 0.004), (0.15, -0.27)	12, 7			
Biomarkers Lymphocytes Lymphocytes [#/v] in Blood by AC		0.052, False, -0.05, [-0.11,0.0]	0.808	16.8k, PASS, (5.6e-04, 0.067), (0.28, 0.15) 61.7k, FAIL, (0.005, 0.621), (0.11, -0.02)	-0.022, 0.02, 0			
Mean Corpuscular Haemoglobin Concentration	[ID: 30060] 130k, 10k, 5%, 20%, 1%	4.3e-04, True, 0.09, [0.04,0.15]	0.495	11.8k, PASS, (0.457, 0.405), (0.07, -0.08)	5, 5			
Biomarkers Mean CP Heme Hemoglobin [M/v] in Blood		0.369, False, 0.03, [-0.03,0.08]	0.59	16.8k, PASS, (0.974, 0.55), (0.0, -0.05) NA	-0.022, 0, 0			
Monocyte Percentage	[ID: 30190] 129k, 10k, 16%, 39%, 9%	1.2e-48, True, 0.43, [0.37,0.48]	0.03	11.7k, PASS, (5.3e-05, 0.001), (0.4, -0.34)	4, 10			
Biomarkers Monocyte % Monocytes [#/v] in Blood by Automated count		2.6e-16, True, -0.24, [-0.3,-0.18]	7.8e-04	16.7k, PASS, (1.9e-04, 9.7e-07), (0.28, -0.35) NA	-0.019, 0, -0.004			
Phosphate	[ID: 30810] 121k, 9k, 9%, 25%, 3%	0.158, False, 0.03, [-0.03,0.08]	0.465	8.6k, PASS, (0.004, 0.385), (0.09, -0.1)	2, 4			
Biomarkers Phosphate Phosphate [Mass/v] in Serum or Plasma		0.085, False, -0.05, [-0.11,0.01]	0.56	15.9k, FAIL, (0.306, 0.749), (0.07, -0.03) 32.2k, PASS, (0.033, 0.013), (0.12, -0.14)	-0.022, 0, 0			
Platelet Crit	[ID: 30090] 130k, 10k, 21%, 52%, 10%	1.1e-50, True, 0.36, [0.3,0.41]	5.8e-04	8.3k, PASS, (4.2e-04, 0.696), (0.42, -0.04)	9, 16			
Biomarkers Platelet Crit Platelets [#/v] in Blood by Automated count		2.7e-16, True, -0.26, [-0.31,-0.2]	2.2e-04	16.8k, PASS, (2.2e-05, 0.123), (0.34, -0.13) NA	-0.022, -0.017, 0			
Pulse Rate	[ID: 4194] 51.2k, 3k, 10%, 39%, 1%	0.728, False, -0.01, [-0.1,0.07]	0.044	3.3k, PASS, (0.523, 0.622), (0.12, 0.07)	2, 1			
Physical Measures Pulse Rate Computed heart rate (mean)		9.2e-04, True, -0.14, [-0.23,-0.06]	0.406	11.7k, PASS, (0.283, 0.773), (-0.1, -0.03) 127k, PASS, (0.337, 0.004), (0.03, -0.08)	-0.034, -0.031, -0.011			
Red Blood Cell Dist Width	[ID: 30070] 129k, 10k, 18%, 41%, 6%	0.006, True, -0.08, [-0.13,-0.02]	4.1e-04	11.8k, FAIL, (0.228, 3.4e-06), (0.11, -0.39)	31, 5			
Biomarkers RBC Dist Width Erythrocyte distribution width [Ratio] by AC		5.7e-60, True, -0.5, [-0.55,-0.44]	0.776	16.6k, PASS, (0.66, 6.9e-06), (0.04, -0.38) 65.1k, PASS, (0.017, 2.3e-10), (0.1, -0.25)	0.005, -0.009, 0			
Red Blood Cell Count	[ID: 30010] 130k, 10k, 19%, 51%, 7%	9.5e-21, True, 0.28, [0.22,0.33]	0.258	11.8k, PASS, (0.002, 0.643), (0.28, -0.04)	13, 13			
Biomarkers RBCs Erythrocytes [#/v] in Blood by AC		0.306, False, 0.03, [-0.03,0.08]	0.005	16.8k, PASS, (0.535, 0.115), (0.25, -0.11) 80.1k, PASS, (1.0e-15, 0.288), (0.3, 0.04)	-0.016, 0.023, -0.006			
Reticulocyte Percentage	[ID: 30240] 127k, 10k, 16%, 40%, 5%	0.004, True, 0.08, [0.02,0.13]	0.263	11.6k, PASS, (0.009, 0.331), (0.24, -0.08)	21, 4			
Biomarkers Reticulocyte % Reticulocytes [#/volume] in Blood		2.9e-05, True, -0.12, [-0.17,-0.06]	0.659	16.4k, PASS, (0.346, 0.419), (0.07, -0.07) NA	-0.015, -0.008, -0.007			
Sodium In Urine	[ID: 30530] 163k, 16k, 5%, 16%, 1%	0.051, False, -0.04, [-0.09,0.01]	0.569	14.2k, FAIL, (0.478, 0.354), (0.06, -0.08)	0, 0			
Biomarkers Sodium In Urine Sodium [Moles/v] in Urine		2.1e-05, True, -0.11, [-0.16,-0.06]	0.087	21.8k, PASS, (0.006, 0.631), (-0.19, 0.03) 3.6k, PASS, (0.761, 0.744), (0.05, 0.05)	-0.012, 0, -0.008			
Testosterone	[ID: 30850] 121k, 9k, 7%, 18%, 2%	0.049, False, -0.06, [-0.11,-0.0]	0.454	9.2k, PASS, (0.474, 0.438), (-0.08, 0.09)	0, 4			
Biomarkers Testosterone Testosterone [M/v] in Serum or Plasma		7.5e-04, True, 0.1, [0.04,0.16]	0.216	15.9k, FAIL, (0.195, 0.019), (0.1, -0.19) 8.6k, PASS, (0.152, 0.008), (0.18, -0.28)	-0.01, -0.009, -0.009			
Total Protein	[ID: 30860] 121k, 9k, 14%, 36%, 3%	0.225, False, 0.04, [-0.02,0.09]	0.078	8.6k, PASS, (0.006, 0.422), (0.09, -0.1)	1, 7			
Biomarkers Total Protein Protein [M/v] in Serum or Plasma		6.7e-05, True, -0.11, [-0.17,-0.06]	0.245	15.9k, PASS, (0.432, 0.06), (0.06, -0.15) 79.9k, PASS, (7.1e-08, 0.335), (0.2, -0.03)	-0.015, -0.009, -0.007			
Triglycerides	[ID: 30870] 133k, 11k, 15%, 41%, 4%	1.5e-07, True, 0.15, [0.09,0.2]	0.21	10.6k, PASS, (0.073, 0.235), (0.18, 0.12)	9, 11			
Biomarkers Triglycerides Triglycerides [M/v] in Serum or Plasma		0.028, True, 0.06, [0.01,0.11]	0.298	17.3k, FAIL, (0.074, 0.512), (0.14, 0.05) 63.1k, PASS, (0.643, 0.655), (0.02, 0.02)	-0.022, 0.015, 0			

Fig. 6: **Trait Table**

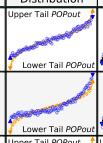
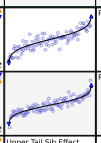
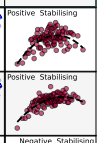
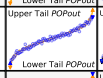
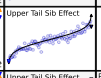
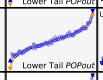
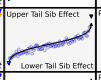
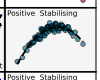
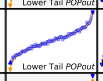
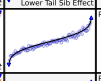
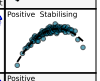
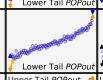
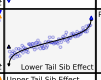
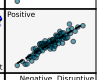
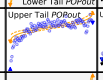
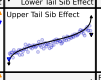
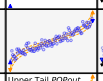
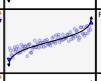
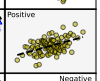
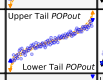
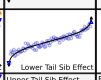
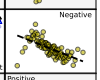
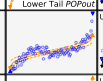
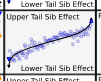
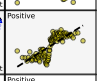
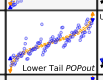
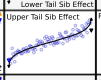
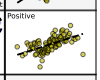
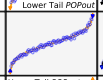
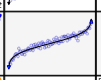
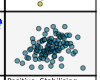
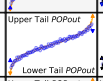
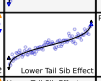
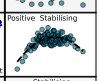
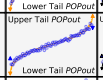
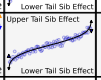
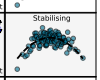
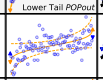
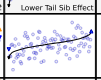
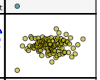
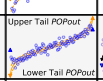
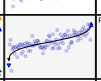
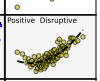
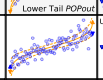
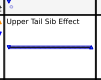
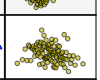
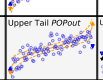
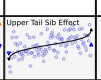
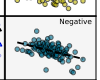
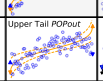
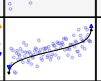
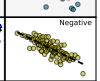
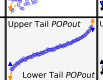
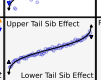
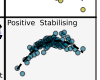
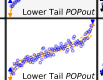
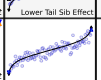
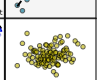
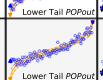
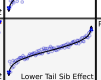
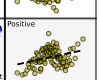
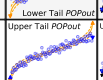
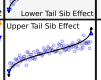
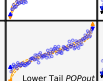
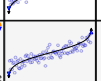
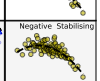
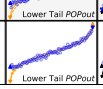
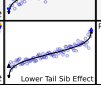
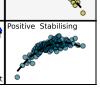
Trait Name	Overview	*POPout* Tests (Upper | Lower)	Sib Tests	Replication	Rare/Evolution	POP (Phenotype on PRS) Distribution	Conditional Sibling Distribution	Selection Inference Distribution
Category, Abbreviated Name (AOU Name)	[UKB ID] Target Size,Sib Pairs SNP-h^2,SIB-h^2,PRS-r^2	Pval., 5% FDR, Effect Size, 95% CI	Meta P-values	Repeated Measures | Non-EUR | All Of Us Target Size, QC, pvals, effect sizes (Lower,Upper)	Rare Variants (Lower, Upper), Model Params: ($Y \sim X_0 + \beta X + \gamma X^2$)			
Urea								
Biomarkers Urea Urea nitrogen [M/v] in Serum or Plasma	[ID: 30670] 133k, 11k, 8%, 27%, 2%	2.4e-06, True, 0.13, [0.07,0.18] / 0.028, True, -0.06, [-0.11,-0.01]	0.201 / 0.662	10.6k, PASS, (0.594, 0.995), (0.05, 0.0) 17.4k, PASS, (0.079, 0.164), (0.12, -0.11) 82.7k, PASS, (0.006, 7.4e-15), (0.1, -0.28)	5, 0 -0.012, 0.01, -0.01			
Vitamin D								
Biomarkers Vitamin D 25-hydroxyvitamin D3 [M/v] in Serum or Plasma	[ID: 30890] 126k, 10k, 7%, 31%, 2%	4.6e-10, True, 0.18, [0.12,0.23] / 0.043, False, 0.07, [0.02,0.12]	0.429 / 0.708	9.8k, PASS, (0.027, 0.828), (0.2, -0.02) 16.2k, PASS, (0.253, 0.435), (0.09, 0.06) 26.8k, PASS, (0.04, 0.123), (0.12, -0.09)	4, 3 -0.01, 0.021, -0.011			
White Blood Cell Count								
Biomarkers WBCs Leukocytes [#/v] in Blood by AC	[ID: 30000] 130k, 10k, 17%, 40%, 4%	0.019, True, 0.06, [0.01,0.12] / 1.0e-05, True, -0.12, [-0.18,-0.07]	0.013 / 0.284	11.8k, PASS, (0.95, 0.291), (0.01, -0.11) 16.8k, PASS, (0.855, 0.344), (0.01, -0.07) 77.8k, PASS, (0.031, 4.2e-09), (0.08, -0.2)	11, 6 -0.017, -0.009, -0.005			
Waist Circumference								
Physical Measures Waist Circum Waist Circumference (Mean)	[ID: 48] 167k, 17k, 14%, 42%, 3%	1.8e-08, True, 0.14, [0.09,0.18] / 0.014, True, 0.06, [0.01,0.1]	0.001 / 0.024	15.5k, PASS, (0.003, 0.286), (0.24, 0.08) 22.5k, PASS, (0.021, 0.018), (0.15, 0.16) NA	1, 0 0.006, 0.032, -0.027			
Weight								
Physical Measures Weight Weight (Mean)	[ID: 21002] 167k, 17k, 20%, 58%, 6%	1.1e-10, True, 0.15, [0.11,0.2] / 0.726, False, 0.01, [-0.04,0.06]	0.488 / 0.497	15.5k, PASS, (0.002, 0.869), (0.25, 0.01) 22.5k, PASS, (2.5e-05, 0.009), (0.26, 0.18) 127k, PASS, (3.2e-04, 5.3e-05), (0.1, -0.11)	4, 1 0.006, 0.035, -0.028			
Hand Grip Strength (left)								
Physical Measures Grip Strength NA	[ID: 46] 167k, 17k, 9%, 31%, 2%	2.1e-14, True, 0.19, [0.14,0.23] / 2.9e-04, True, 0.09, [0.04,0.14]	0.064 / 5.5e-04	15.4k, PASS, (0.138, 0.478), (0.13, -0.05) 22.5k, PASS, (0.087, 2.7e-04), (-0.1, -0.23) NA	0, 6 -0.021, 0.045, 0			
Age Completed Full Time Education								
Questionnaire/Cognitive Education Age NA	[ID: 845] 109k, 9k, 5%, 65%, 1%	4.0e-10, True, -0.23, [-0.32,-0.15] / 9.9e-07, True, -0.15, [-0.21,-0.09]	4.3e-06 / 0.074	8k, PASS, (0.006, 0.197), (-0.38, -0.14) 11.4k, FAIL, (0.12, 1.0e-05), (-0.14, -0.4) NA	0, 0 -0.022, -0.044, 0.003			
Time Spend Outdoors In Summer								
Questionnaire/Cognitive Outdoor Time NA	[ID: 1050] 152k, 14k, 5%, 29%, 1%	0.118, False, 0.04, [-0.01,0.09] / 0.056, False, -0.05, [-0.1,0.0]	0.265 / 0.376	13k, PASS, (0.123, 0.831), (0.12, 0.02) 18.6k, FAIL, (8.9e-13, 0.044), (0.53, 0.14) NA	0, 0 -0.016, 0.01, 0			
Time Spent Watching Television								
Questionnaire/Cognitive TV Watching NA	[ID: 1070] 158k, 15k, 8%, 45%, 1%	8.3e-04, True, 0.09, [0.04,0.14] / 0.01, True, -0.06, [-0.11,-0.01]	0.061 / 0.042	14k, PASS, (0.708, 0.544), (0.03, -0.06) 20.7k, PASS, (1.8e-04, 0.805), (0.26, -0.02) NA	0, 0 -0.026, -0.023, 0			
Age Started Wearing Glasses Or Contact Lenses								
Questionnaire/Cognitive Glasses Age NA	[ID: 2217] 143k, 13k, 6%, 30%, 1%	0.51, False, 0.01, [-0.03,0.06] / 0.044, False, 0.09, [0.01,0.17]	0.035 / 0.016	12.9k, FAIL, (0.272, 0.023), (-0.07, 0.36) 17k, PASS, (0.089, 0.47), (0.14, -0.05) NA	0, 2 -0.018, 0.054, 0			
Age When Periods Started (menarche)								
Questionnaire/Cognitive Menarche Age NA	[ID: 2714] 85.6k, 5k, 18%, 51%, 3%	1.3e-07, True, 0.18, [0.11,0.25] / 0.052, False, -0.07, [-0.14,0.0]	0.015 / 0.054	7.4k, PASS, (0.013, 0.348), (0.26, -0.11) 11.2k, FAIL, (0.003, 0.244), (0.28, 0.11) NA	0, 0 -0.014, 0.023, 0			
Ankle Spacing Width								
Physical Measures Ankle Spacing NA	[ID: 3143] 96k, 7k, 23%, 65%, 6%	0.168, False, 0.04, [-0.02,0.11] / 0.215, False, -0.04, [-0.1,0.02]	0.371 / 0.873	NA 8.3k, FAIL, (0.048, 0.876), (0.23, -0.02) NA	3, 0 -0.022, 0, 0			
Heel Broadband Ultrasound Attenuation								
Physical Measures Heel BUA NA	[ID: 3144] 95.9k, 7k, 22%, 51%, 7%	1.2e-16, True, 0.29, [0.22,0.36] / 3.1e-21, True, -0.34, [-0.4,-0.27]	0.08 / 0.019	NA 8.3k, PASS, (0.192, 1.1e-07), (0.13, -0.47) NA	1, 2 0.003, 0.01, -0.014			
Heel Bone Mineral Density								
Physical Measures Heel BMD NA	[ID: 3148] 95.8k, 7k, 24%, 58%, 9%	2.7e-27, True, 0.39, [0.32,0.46] / 2.5e-16, True, -0.29, [-0.36,-0.22]	0.003 / 0.002	NA 8.3k, PASS, (0.006, 3.7e-06), (0.25, -0.47) NA	2, 3 0.003, 0, -0.014			
Age Started Hormone-replacement Therapy (hrt)								
Questionnaire/Cognitive HRT Age NA	[ID: 3536] 50.1k, 1024, 6%, 21%, 1%	0.327, False, 0.06, [-0.06,0.17] / 0.441, False, -0.05, [-0.17,0.07]	0.707 / 0.327	2.5k, PASS, (0.386, 0.555), (-0.15, -0.13) 2.1k, PASS, (0.978, 0.642), (-0.0, -0.1) NA	0, 0 -0.019, 0, 0			
Age At Menopause								
Questionnaire/Cognitive Menopause Age NA	[ID: 3581] 59.4k, 2k, 12%, 35%, 2%	1.1e-14, True, 0.36, [0.27,0.45] / 0.035, True, -0.1, [-0.19,-0.02]	0.623 / 0.821	4.2k, PASS, (0.062, 0.73), (0.33, -0.05) 5.3k, PASS, (0.259, 0.819), (0.17, -0.03) NA	1, 1 -0.044, 0.041, 0.013			
Age Hay Fever								
Questionnaire/Cognitive Age Hay Fever NA	[ID: 3761] 53.9k, 864, 8%, 32%, 1%	0.6, False, 0.03, [-0.08,0.14] / 0.058, False, -0.09, [-0.19,0.0]	0.043 / 0.107	2.5k, PASS, (0.505, 0.883), (-0.1, -0.04) 4.9k, PASS, (0.316, 0.147), (-0.16, -0.21) NA	0, 1 -0.042, 0, 0			
Position Of Pulse Wave Notch								
Physical Measures PW Notch NA	[ID: 4199] 55.3k, 3k, 6%, 17%, 1%	0.74, False, 0.01, [-0.07,0.09] / 1.1e-04, True, -0.16, [-0.24,-0.08]	0.025 / 0.272	3.6k, PASS, (0.911, 0.292), (-0.02, 0.15) 12.5k, PASS, (0.82, 0.94), (0.02, 0.01) NA	0, 0 -0.044, -0.017, 0			
Time To Answer								
Questionnaire/Cognitive Time To Answer NA	[ID: 4288] 55.7k, 3k, 6%, 26%, 1%	0.364, False, 0.04, [-0.04,0.12] / 3.4e-04, True, -0.16, [-0.24,-0.07]	0.634 / 0.33	3.6k, PASS, (0.196, 0.464), (-0.23, -0.15) 12.3k, PASS, (0.455, 0.941), (-0.07, -0.01) NA	0, 0 -0.044, -0.04, 0			
Sitting Height								
Physical Measures Sitting Height NA	[ID: 20015] 167k, 17k, 29%, 74%, 13%	3.5e-60, True, 0.44, [0.39,0.49] / 4.3e-25, True, -0.27, [-0.32,-0.22]	3.3e-08 / 7.1e-04	15.1k, PASS, (0.009, 0.923), (0.21, -0.11) 22.4k, PASS, (6.6e-06, 1.5e-04), (0.31, -0.28) NA	10, 23 -0.008, 0.015, -0.013			
Fluid Intelligence Score								
Questionnaire/Cognitive Fluid Intellect NA	[ID: 20016] 54.5k, 3k, 17%, 60%, 2%	7.7e-04, True, 0.14, [0.06,0.22] / 0.511, False, 0.03, [-0.05,0.1]	0.556 / 0.67	3.6k, PASS, (0.53, 0.218), (-0.1, -0.24) 10.3k, PASS, (0.565, 0.02), (-0.06, 0.25) NA	0, 0 -0.043, 0, 0			
Birth Weight								
Questionnaire/Cognitive Birth Weight NA	[ID: 20022] 95.6k, 6k, 8%, 70%, 1%	2.2e-07, True, 0.17, [0.11,0.23] / 0.336, False, 0.03, [-0.03,0.1]	0.088 / 0.038	7.1k, FAIL, (0.002, 0.354), (0.4, 0.11) 6.7k, FAIL, (0.356, 0.069), (-0.12, -0.18) NA	2, 0 -0.021, 0.011, 0			
Mean Time To Correctly Identify Matches								
Questionnaire/Cognitive Match ID Time NA	[ID: 20023] 166k, 16k, 6%, 36%, 1%	0.677, False, 0.01, [-0.04,0.06] / 9.9e-07, True, -0.12, [-0.17,-0.07]	0.043 / 0.164	15.4k, PASS, (0.411, 0.005), (-0.07, -0.24) 21.2k, FAIL, (6.0e-06, 0.005), (0.3, 0.2) NA	1, 0 -0.013, -0.026, -0.007			
Neuroticism Score								
Questionnaire/Cognitive Neuroticism NA	[ID: 20127] 136k, 11k, 8%, 28%, 1%	7.0e-07, True, 0.14, [0.08,0.19] / 0.334, False, -0.03, [-0.08,0.03]	0.297 / 0.422	NA 14.8k, PASS, (1.1e-04, 0.104), (0.3, -0.15) NA	0, 0 -0.006, -0.02, -0.011			
Forced Expiratory Volume In 1-second Best Measure								
Physical Measures FEV Best NA	[ID: 20150] 126k, 9k, 17%, 48%, 5%	7.8e-21, True, 0.27, [0.21,0.33] / 0.327, False, -0.03, [-0.08,0.03]	0.283 / 0.024	NA 742, PASS, (0.541, 0.795), (0.16, 0.08) NA	1, 4 0.0, 0.023, -0.011			

Fig. 7: **Trait Table**

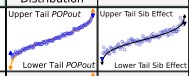
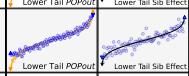
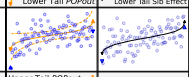
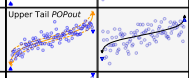
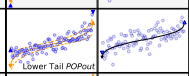
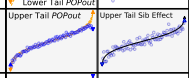
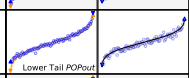
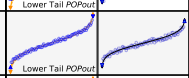
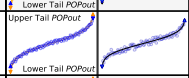
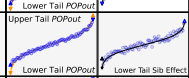
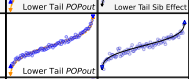
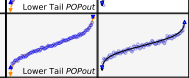
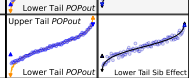
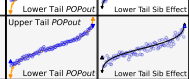
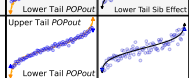
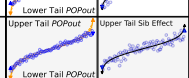
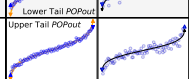
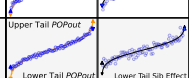
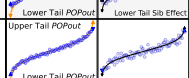
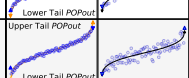
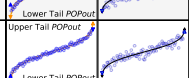
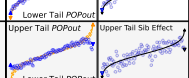
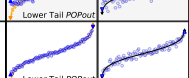
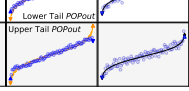
Trait Name	Overview	*POPout* Tests (Upper | Lower)	Sib Tests	Replication	Rare/Evolution	POP (Phenotype on PRS) Distribution	Conditional Sibling Distribution	Selection Inference Distribution
Category, Abbreviated Name (AOU Name)	[UKB ID] Target Size,Sib Pairs SNP-h^2,SIB-h^2,PRS-r^2	Pval., 5% FDR, Effect Size, 95% CI	Meta P-values	Repeated Measures | Non-EUR | All Of Us Target Size, QC, pvals, effect sizes (Lower,Upper)	Rare Variants (Lower, Upper), Model Params: ($Y \sim X\alpha + \beta X + \gamma X^2$)			
Forced Vital Capacity Best Measure Physical Measures FVC Best NA	[ID: 20151] 126k, 9k, 20%, 55%, 6%	9.4e-27, True, 0.32, [0.26,0.38] 0.02, True, -0.07, [-0.12,-0.01]	0.006 0.036	NA 742, PASS, (0.763, 0.583), (-0.09, -0.06) NA	1, 3 0.002, 0.026, -0.013			
Forced Expiratory Volume In 1-second Predicted Percentage Physical Measures Pred FEV % NA	[ID: 20154] 98.2k, 6k, 16%, 38%, 3%	2.7e-12, True, 0.24, [0.17,0.31] 0.39, False, -0.03, [-0.09,0.04]	0.291 0.043	NA 569, PASS, (0.152, 0.268), (0.41, 0.37) NA	0, 0 -0.004, 0.015, 0			
Duration To Complete Numeric Path (trail #1) Questionnaire/Cognitive Numeric Trail NA	[ID: 20156] 56.7k, 1k, 5%, 31%, 1%	0.027, True, -0.11, [-0.21,-0.01] 0.206, False, -0.06, [-0.16,0.04]	0.554 0.284	NA 2.3k, PASS, (0.005, 0.261), (0.73, 0.27) NA	0, 0 -0.062, 0, 0			
Duration To Complete Alphanumeric Path (trail #2) Questionnaire/Cognitive Alphanum. Trail NA	[ID: 20157] 56.7k, 1k, 11%, 47%, 1%	0.092, False, -0.08, [-0.18,0.01] 1.3e-04, True, -0.19, [-0.29,-0.1]	0.2 0.934	NA 2.3k, PASS, (1.0e-07, 0.026), (-1.07, -0.55) NA	0, 0 -0.062, -0.013, 0			
Number Of Symbol Digit Matches Made Correctly Questionnaire/Cognitive SD Matches NA	[ID: 20159] 51.6k, 1k, 11%, 39%, 1%	0.003, True, 0.15, [0.05,0.25] 0.01, True, 0.12, [0.03,0.21]	0.779 0.522	NA 2.8k, PASS, (0.832, 0.209), (0.04, 0.21) NA	0, 0 -0.054, 0, 0			
Forced Expiratory Volume In 1-second Z-score Physical Measures FEV Z-score NA	[ID: 20256] 134k, 11k, 15%, 40%, 4%	0.038, False, 0.06, [0.0,0.11] 2.2e-26, True, -0.29, [-0.35,-0.24]	0.025 0.655	NA 811, PASS, (0.501, 0.72), (0.24, -0.08) NA	1, 0 -0.007, -0.018, -0.006			
Body Fat Percentage Physical Measures Body Fat % NA	[ID: 23099] 165k, 16k, 17%, 49%, 5%	1.1e-12, True, 0.18, [0.13,0.23] 0.016, True, 0.06, [0.01,0.1]	0.199 0.376	15.1k, PASS, (0.022, 0.497), (0.19, 0.05) 22.2k, PASS, (0.019, 0.991), (0.16, 0.0) NA	1, 0 0.005, 0, -0.025			
Basal Metabolic Rate Physical Measures BMR NA	[ID: 23105] 165k, 16k, 23%, 68%, 9%	1.1e-04, True, 0.1, [0.05,0.14] 0.211, False, -0.03, [-0.07,0.02]	0.139 0.249	15.1k, PASS, (0.415, 0.759), (0.07, -0.02) 22.2k, PASS, (2.9e-04, 5.1e-07), (0.28, 0.3) NA	5, 10 0.002, 0.042, -0.023			
Impedance Of Whole Body Physical Measures Body Impedance NA	[ID: 23106] 165k, 16k, 21%, 54%, 7%	5.2e-06, True, 0.11, [0.07,0.16] 0.003, True, -0.07, [-0.12,-0.03]	0.348 0.38	15.1k, PASS, (0.372, 0.282), (0.08, 0.09) 22.2k, PASS, (0.518, 0.743), (-0.04, 0.02) NA	5, 10 -0.004, -0.045, -0.015			
Right Impedance Of Leg Physical Measures Leg Impedance NA	[ID: 23107] 165k, 16k, 19%, 54%, 6%	5.2e-11, True, 0.17, [0.12,0.22] 0.009, True, -0.06, [-0.11,-0.02]	0.603 8.5e-04	15.1k, PASS, (0.007, 0.357), (0.23, 0.08) 22.2k, PASS, (0.053, 0.817), (0.13, -0.02) NA	4, 4 -0.007, -0.022, -0.013			
Left Leg Fat Percentage Physical Measures Leg Fat % NA	[ID: 23115] 165k, 16k, 16%, 48%, 4%	1.7e-09, True, 0.15, [0.1,0.19] 0.462, False, 0.02, [-0.03,0.06]	0.647 0.335	15.1k, PASS, (0.013, 0.514), (0.2, 0.05) 22.2k, PASS, (0.032, 0.797), (0.14, -0.02) NA	1, 0 0.005, 0.027, -0.025			
Trunk Fat-free Mass Physical Measures FF Trunk Mass NA	[ID: 23129] 165k, 16k, 25%, 71%, 10%	8.7e-14, True, 0.18, [0.13,0.23] 0.417, False, 0.02, [-0.03,0.06]	0.105 0.088	15.1k, PASS, (0.331, 0.152), (0.08, -0.12) 22.2k, PASS, (0.063, 0.014), (0.13, 0.15) NA	4, 14 -0.002, 0.047, -0.019			
Mean Corpuscular Volume Biomarkers Mean CP Vol NA	[ID: 30040] 130k, 10k, 23%, 54%, 11%	3.0e-97, True, 0.44, [0.34,0.55] 2.0e-05, True, -0.11, [-0.16,-0.05]	0.309 5.1e-05	11.8k, FAIL, (1.0e-07, 0.04), (0.51, -0.2) 16.7k, FAIL, (0.82, 0.008), (-0.02, 0.21) NA	15, 21 -0.022, -0.03, 0			
Monocyte Count Biomarkers Monocytes Monocytes [#/v] in Blood by Automated count	[ID: 30130] 129k, 10k, 18%, 45%, 7%	1.8e-28, True, 0.33, [0.27,0.39] 7.8e-06, True, -0.13, [-0.19,-0.07]	0.387 0.002	11.7k, PASS, (0.007, 0.015), (0.26, -0.23) 16.7k, PASS, (4.5e-04, 0.703), (0.28, -0.03) 59.9k, FAIL, (0.325, 0.141), (0.04, -0.06)	6, 6 -0.022, 0, 0			
Neutrophill Count Biomarkers Neutrophills Neutrophils [#/v] in Blood by Automated count	[ID: 30140] 129k, 10k, 14%, 35%, 4%	1.7e-08, True, 0.16, [0.11,0.22] 9.6e-11, True, -0.18, [-0.24,-0.13]	0.084 0.401	11.7k, PASS, (0.7, 0.398), (0.04, -0.08) 16.7k, PASS, (0.226, 0.274), (-0.1, -0.08) 63.5k, FAIL, (0.007, 0.53), (-0.11, -0.02)	9, 7 -0.016, -0.021, -0.007			
Neutrophill Percentage Biomarkers Neutrophill % Neutrophils/100 leukocytes in Blood by AC	[ID: 30200] 129k, 10k, 12%, 32%, 3%	3.1e-06, True, 0.13, [0.08,0.19] 5.6e-13, True, -0.21, [-0.26,-0.15]	0.045 0.2	11.7k, PASS, (0.131, 5.2e-04), (0.15, -0.32) 16.8k, PASS, (0.093, 0.002), (0.13, -0.25) 53k, PASS, (0.01, 3.0e-05), (0.11, -0.19)	7, 8 -0.014, -0.031, -0.008			
Reticulocyte Count Biomarkers Reticulocytes NA	[ID: 30250] 127k, 10k, 16%, 41%, 5%	0.046, False, 0.06, [0.0,0.11] 1.6e-04, True, -0.11, [-0.16,-0.05]	0.18 0.529	11.6k, PASS, (0.01, 0.699), (0.23, -0.03) 16.4k, PASS, (0.198, 0.307), (0.1, -0.09) NA	21, 8 -0.014, 0.015, -0.008			
Mean Reticulocyte Volume Biomarkers Mean RT Vol NA	[ID: 30260] 128k, 10k, 17%, 45%, 8%	1.4e-21, True, 0.28, [0.22,0.34] 1.7e-08, True, -0.17, [-0.23,-0.11]	0.337 0.014	11.6k, PASS, (0.213, 0.045), (0.12, -0.21) 16.4k, PASS, (0.732, 0.045), (-0.03, -0.16) NA	15, 20 -0.015, 0.014, -0.008			
Mean Sphered Cell Volume Biomarkers Sphered Cell Vol NA	[ID: 30270] 128k, 10k, 18%, 47%, 8%	8.6e-06, True, 0.13, [0.07,0.18] 1.1e-08, True, -0.16, [-0.21,-0.11]	0.21 0.066	8.2k, PASS, (0.11, 0.011), (0.13, -0.28) 16.4k, FAIL, (0.054, 0.063), (0.16, 0.15) NA	16, 17 -0.016, -0.032, -0.006			
Immature Reticulocyte Fraction Biomarkers Imm. RT Frac NA	[ID: 30280] 128k, 10k, 11%, 31%, 4%	1.5e-15, True, 0.21, [0.15,0.26] 4.7e-04, True, -0.1, [-0.15,-0.04]	0.914 0.541	8.2k, PASS, (0.357, 0.894), (0.1, 0.02) 16.4k, PASS, (0.023, 0.066), (0.17, -0.14) NA	10, 3 -0.022, 0, 0			
Apolipoprotein A Biomarkers Apolipoprotein A NA	[ID: 30630] 121k, 9k, 17%, 43%, 8%	2.9e-04, True, 0.1, [0.05,0.16] 5.5e-06, True, -0.13, [-0.18,-0.07]	0.263 0.357	8.5k, PASS, (0.048, 0.658), (0.22, 0.05) 15.9k, PASS, (0.957, 0.007), (-0.0, -0.22) NA	12, 14 -0.012, -0.025, -0.009			
Glucose Biomarkers Glucose NA	[ID: 30740] 121k, 9k, 6%, 22%, 2%	0.026, True, 0.07, [0.03,0.11] 1.1e-13, True, -0.22, [-0.28,-0.16]	0.005 0.439	8.6k, PASS, (0.02, 0.002), (0.27, -0.37) 15.8k, PASS, (0.007, 0.136), (0.21, -0.12) NA	2, 3 -0.022, -0.015, 0			
Igf-1 Biomarkers IGF-1 NA	[ID: 30770] 132k, 11k, 20%, 49%, 6%	4.4e-09, True, 0.16, [0.11,0.22] 0.343, False, -0.02, [-0.08,0.03]	0.419 0.219	10.5k, PASS, (0.319, 0.781), (0.09, -0.03) 17.3k, PASS, (0.142, 0.038), (0.11, -0.16) NA	10, 12 -0.012, -0.008, -0.01			
Shbg Biomarkers SHBG NA	[ID: 30830] 120k, 9k, 16%, 50%, 8%	0.07, False, 0.05, [-0.0,0.1] 1.1e-32, True, -0.36, [-0.42,-0.3]	0.15 0.506	8.4k, PASS, (0.028, 0.004), (0.28, -0.3) 15.8k, PASS, (0.967, 0.002), (0.0, -0.26) NA	6, 8 -0.011, -0.025, -0.011			

Fig. 8: **Trait Table**